

C v3 Vertical Slice v1 - Stage 2 Human Semantic Review

Renderer-only artifact from the frozen existing Stage 2 output. No Stage 2 execution is performed by this document builder.

Frozen Stage 2 source SHA-256: d70c437a536d21765a91693fd8e292b47cb3f157f89b503b135e52b974b74d86

LI4 - RESOLVED

WHO primary Location

S:LI4:location: On the dorsum of the hand, radial to the midpoint of the second metacarpal bone.

Semantic decomposition / operands

LM:S:LI4:location:36-79
source: the midpoint of the second metacarpal bone
class: bone.bone | status: RESOLVED | executor: fma_mesh
identity: {"fma_id":"FMA23900","kind":"fma_concept","part_ids":["FJ3243","FJ3352"]}
reason: None

LM:S:LI4:location:53-79
source: the second metacarpal bone
class: bone.bone | status: RESOLVED | executor: fma_mesh
identity: {"fma_id":"FMA23900","kind":"fma_concept","part_ids":["FJ3243","FJ3352"]}
reason: None

LM:S:LI4:location:7-25
source: dorsum of the hand
class: surface.aspect | status: RESOLVED | executor: approved_surface_registry
identity: {"geometry_hash":"660965ac347a4b5e19c1f02c3d45a8be59be919076604c2a490ce1a7242bf5f4","kind":"surface_registry","registry_id":"SR:dorsum_hand"}
reason: None

Relation / operator chain

RL:S:LI4:location:00:relative-to
status: RESOLVED | operator: directional_relation
operand IDs: LM:S:LI4:location:36-79
direction: radial | source statement: S:LI4:location
reason: None

RL:S:LI4:location:01:midpoint-of-entity
status: RESOLVED | operator: entity_midpoint
operand IDs: LM:S:LI4:location:53-79
direction: None | source statement: S:LI4:location
reason: None

RL:S:LI4:location:02:surface-landmark
status: RESOLVED | operator: surface_membership
operand IDs: LM:S:LI4:location:7-25
direction: None | source statement: S:LI4:location
reason: None

Dependency edges (projection of existing explicit IDs)

LM:S:LI4:location:36-79 -> RL:S:LI4:location:00:relative-to [operand_to_relation]
LM:S:LI4:location:53-79 -> RL:S:LI4:location:01:midpoint-of-entity [operand_to_relation]
LM:S:LI4:location:7-25 -> RL:S:LI4:location:02:surface-landmark [operand_to_relation]
RL:S:LI4:location:00:relative-to -> S:LI4:location [relation_to_statement]
RL:S:LI4:location:01:midpoint-of-entity -> S:LI4:location [relation_to_statement]
RL:S:LI4:location:02:surface-landmark -> S:LI4:location [relation_to_statement]
S:LI4:location -> LI4 [statement_to_point]

Recorded data order

Stage 2 output does not record timestamped per-operation runtime order. The following preserves exact serialized Stage 2 data order.

LM:S:LI4:location:36-79 -> LM:S:LI4:location:53-79 -> LM:S:LI4:location:7-25 -> RL:S:LI4:location:00:relative-to -> RL:S:LI4:location:01:midpoint-of-entity -> RL:S:LI4:location:02:surface-landmark -> S:LI4:location -> LI4

Final semantic execution result

{"location_statement_status":"RESOLVED","primary_location_status":"RESOLVED"}

Coordinates generated: 0 | Legacy C coordinate referenced: 0

- [] semantic decomposition correct
- [] operator semantics correct
- [] operand identity correct
- [] dependency structure correct
- [] unresolved/conditional handling correct
- [] ACCEPT
- [] REJECT

GB23 - RESOLVED

WHO primary Location

S:GB23:location: In the lateral thoracic region, in the fourth intercostal space, 1 B-cun anterior to the midaxillary line.

Semantic decomposition / operands

LM:S:GB23:location:39-63
source: fourth intercostal space
class: space.anatomical_space | status: RESOLVED | executor: approved_surface_registry
identity: {"geometry_hash":"c8e8691e4f765638eb499518792d13bea202f0a5aa3a1f90a050c51fb2e2ed36","kind":"surface_registry","registry_id":"SR:fourth_intercostal_space"}

LM:S:GB23:location:7-30
source: lateral thoracic region
class: region.body_region | status: RESOLVED | executor: approved_surface_registry
identity: {"geometry_hash":"8cf739f6e831150ec0c56595886dd8d46bc412a09ff8ff132ad1ae55e653d9da","kind":"surface_registry","registry_id":"SR:lateral_thorax"}

LM:S:GB23:location:84-105
source: the midaxillary line
class: line.anatomical_line | status: RESOLVED | executor: approved_surface_registry
identity: {"geometry_hash":"643f43e0388878b8eb9f6175a1c685aeb1b7b80df0274d8adc5b0b25cfda1420","kind":"surface_registry","registry_id":"SR:midaxillary_line"}

Relation / operator chain

RL:S:GB23:location:00:relative-to
status: RESOLVED | operator: directional_relation
operand IDs: LM:S:GB23:location:84-105
direction: anterior | source statement: S:GB23:location
reason: None

RL:S:GB23:location:01:surface-landmark
status: RESOLVED | operator: surface_membership
operand IDs: LM:S:GB23:location:7-30
direction: None | source statement: S:GB23:location
reason: None

RL:S:GB23:location:02:surface-landmark
status: RESOLVED | operator: surface_membership
operand IDs: LM:S:GB23:location:39-63
direction: None | source statement: S:GB23:location
reason: None

Measurements / calibration use

{"anchor_landmark_id":"LM:S:GB23:location:84-105","direction":"anterior","executor":"approved_calibration","kind":"proportional_offset","measurement_id":"PM:S:GB23:location:00","reason":null,"source_statement_id":"S:GB23:location","source_unit":"B-cun","status":"RESOLVED","unit":"B-cun","value":1.0,"who_pdf_page":192}

Dependency edges (projection of existing explicit IDs)

LM:S:GB23:location:84-105 -> PM:S:GB23:location:00 [anchor_to_measurement]
PM:S:GB23:location:00 -> S:GB23:location [measurement_to_statement]
LM:S:GB23:location:39-63 -> RL:S:GB23:location:02:surface-landmark [operand_to_relation]
LM:S:GB23:location:7-30 -> RL:S:GB23:location:01:surface-landmark [operand_to_relation]
LM:S:GB23:location:84-105 -> RL:S:GB23:location:00:relative-to [operand_to_relation]
RL:S:GB23:location:00:relative-to -> S:GB23:location [relation_to_statement]
RL:S:GB23:location:01:surface-landmark -> S:GB23:location [relation_to_statement]
RL:S:GB23:location:02:surface-landmark -> S:GB23:location [relation_to_statement]
S:GB23:location -> GB23 [statement_to_point]

Recorded data order

Stage 2 output does not record timestamped per-operation runtime order. The following preserves exact serialized Stage 2 data order.

LM:S:GB23:location:39-63 -> LM:S:GB23:location:7-30 -> LM:S:GB23:location:84-105 -> PM:S:GB23:location:00 -> RL:S:GB23:location:00:relative-to -> RL:S:GB23:location:01:surface-landmark -> RL:S:GB23:location:02:surface-landmark -> S:GB23:location -> GB23

Final semantic execution result

{"location_statement_status":"RESOLVED","primary_location_status":"RESOLVED"}

Coordinates generated: 0 | Legacy C coordinate referenced: 0

- ☐ semantic decomposition correct
- ☐ operator semantics correct
- ☐ operand identity correct
- ☐ dependency structure correct
- ☐ unresolved/conditional handling correct
- ☐ ACCEPT
- ☐ REJECT

ST10 - RESOLVED

WHO primary Location

S:ST10:location: In the anterior region of the neck, at the same level as the cricoid cartilage, just anterior to the border of the sternocleido-mastoid muscle.

Semantic decomposition / operands

LM:S:ST10:location:100-107
source: border
class: boundary.border | status: UNRESOLVED | executor: no_family_executor
identity: {"kind":null}
reason: unsupported landmark disposition/class: geometry_or_surface/boundary.border

LM:S:ST10:location:111-142
source: the sternocleido-mastoid muscle
class: muscle | status: RESOLVED | executor: fma_mesh
identity: {"fma_id":"FMA13407","kind":"fma_concept","part_ids":["FJ1573","FJ1595"]}
reason: None

LM:S:ST10:location:57-78
source: the cricoid cartilage
class: cartilage | status: RESOLVED | executor: fma_mesh
identity: {"fma_id":"FMA9615","kind":"fma_concept","part_ids":["FJ2440","FJ2769"]}
reason: None

LM:S:ST10:location:7-34
source: anterior region of the neck
class: region.body_region | status: RESOLVED | executor: approved_surface_registry
identity: {"geometry_hash":"71f7a5ab627b3c93a9aceb574aabdb27ca3577688316b7de105a5effa7840dc7","kind":"surface_registry","registry_id":"SR:anterior_neck"}
reason: None

LM:S:ST10:location:96-142
source: the border of the sternocleido-mastoid muscle
class: muscle | status: RESOLVED | executor: fma_mesh
identity: {"fma_id":"FMA13407","kind":"fma_concept","part_ids":["FJ1573","FJ1595"]}
reason: None

Relation / operator chain

RL:S:ST10:location:00:relative-to
status: RESOLVED | operator: directional_relation
operand IDs: LM:S:ST10:location:96-142
direction: anterior | source statement: S:ST10:location
reason: None

RL:S:ST10:location:01:same-level
status: RESOLVED | operator: same_level_plane
operand IDs: LM:S:ST10:location:57-78
direction: None | source statement: S:ST10:location
reason: None

RL:S:ST10:location:02:surface-landmark
status: RESOLVED | operator: surface_membership
operand IDs: LM:S:ST10:location:7-34
direction: None | source statement: S:ST10:location
reason: None

Dependency edges (projection of existing explicit IDs)

LM:S:ST10:location:57-78 -> RL:S:ST10:location:01:same-level [operand_to_relation]
LM:S:ST10:location:7-34 -> RL:S:ST10:location:02:surface-landmark [operand_to_relation]
LM:S:ST10:location:96-142 -> RL:S:ST10:location:00:relative-to [operand_to_relation]
RL:S:ST10:location:00:relative-to -> S:ST10:location [relation_to_statement]
RL:S:ST10:location:01:same-level -> S:ST10:location [relation_to_statement]
RL:S:ST10:location:02:surface-landmark -> S:ST10:location [relation_to_statement]
S:ST10:location -> ST10 [statement_to_point]

Recorded data order

Stage 2 output does not record timestamped per-operation runtime order. The following preserves exact serialized Stage 2 data order.

LM:S:ST10:location:100-107 -> LM:S:ST10:location:111-142 -> LM:S:ST10:location:57-78 -> LM:S:ST10:location:7-34 -> LM:S:ST10:location:96-142 -> RL:S:ST10:location:00:relative-to -> RL:S:ST10:location:01:same-level -> RL:S:ST10:location:02:surface-landmark -> S:ST10:location -> ST10

Final semantic execution result

{"location_statement_status":"RESOLVED","primary_location_status":"RESOLVED"}

Coordinates generated: 0 | Legacy C coordinate referenced: 0

- ☐ semantic decomposition correct
- ☐ operator semantics correct
- ☐ operand identity correct
- ☐ dependency structure correct
- ☐ unresolved/conditional handling correct
- ☐ ACCEPT
- ☐ REJECT

LI18 - RESOLVED

WHO primary Location

S:LI18:location: On the anterior aspect of the neck, at the same level as the superior border of the thyroid cartilage, between the anterior and posterior borders of the sternocleidomas-toid muscle.

Semantic decomposition / operands

LM:S:LI18:location:115-180:0f3e9745
source: anterior and posterior borders of the sternocleidomas-toid muscle
class: muscle | status: RESOLVED | executor: fma_mesh
identity: {"fma_id":"FMA13407","kind":"fma_concept","part_ids":["FJ1573","FJ1595"]}
reason: None

LM:S:LI18:location:128-180:9046d75e
source: posterior borders of the sternocleidomas-toid muscle
class: muscle | status: RESOLVED | executor: fma_mesh
identity: {"fma_id":"FMA13407","kind":"fma_concept","part_ids":["FJ1573","FJ1595"]}
reason: None

LM:S:LI18:location:57-101
source: the superior border of the thyroid cartilage
class: cartilage | status: RESOLVED | executor: fma_mesh
identity: {"fma_id":"FMA55099","kind":"fma_concept","part_ids":["FJ2808"]}
reason: None

LM:S:LI18:location:61-76
source: superior border
class: boundary.border | status: RESOLVED | executor: composite_subfeature
identity: {"kind":"entity_subfeature","modifier":"superior","parent_landmark_id":"LM:S:LI18:location:84-101"}
reason: None

LM:S:LI18:location:7-22
source: anterior aspect
class: surface.aspect | status: RESOLVED | executor: approved_surface_registry
identity: {"geometry_hash":"71f7a5ab627b3c93a9aceb574aabdb27ca3577688316b7de105a5effa7840dc7","kind":"surface_registry","registry_id":"SR:anterior_neck"}
reason: None

LM:S:LI18:location:84-101
source: thyroid cartilage
class: cartilage | status: RESOLVED | executor: fma_mesh
identity: {"fma_id":"FMA55099","kind":"fma_concept","part_ids":["FJ2808"]}
reason: None

Relation / operator chain

RL:S:LI18:location:00:between
status: RESOLVED | operator: between_constraint
operand IDs: LM:S:LI18:location:115-180:0f3e9745, LM:S:LI18:location:128-180:9046d75e
direction: None | source statement: S:LI18:location
reason: None

RL:S:LI18:location:01:same-level
status: RESOLVED | operator: same_level_plane
operand IDs: LM:S:LI18:location:57-101
direction: None | source statement: S:LI18:location
reason: None

RL:S:LI18:location:02:surface-landmark
status: RESOLVED | operator: surface_membership
operand IDs: LM:S:LI18:location:7-22
direction: None | source statement: S:LI18:location
reason: None

Dependency edges (projection of existing explicit IDs)

LM:S:LI18:location:115-180:0f3e9745 -> RL:S:LI18:location:00:between [operand_to_relation]

LM:S:LI18:location:128-180:9046d75e -> RL:S:LI18:location:00:between [operand_to_relation]

LM:S:LI18:location:57-101 -> RL:S:LI18:location:01:same-level [operand_to_relation]

LM:S:LI18:location:7-22 -> RL:S:LI18:location:02:surface-landmark [operand_to_relation]

RL:S:LI18:location:00:between -> S:LI18:location [relation_to_statement]

RL:S:LI18:location:01:same-level -> S:LI18:location [relation_to_statement]

RL:S:LI18:location:02:surface-landmark -> S:LI18:location [relation_to_statement]

S:LI18:location -> LI18 [statement_to_point]

Recorded data order

Stage 2 output does not record timestamped per-operation runtime order. The following preserves exact serialized Stage 2 data order.

LM:S:LI18:location:115-180:0f3e9745 -> LM:S:LI18:location:128-180:9046d75e -> LM:S:LI18:location:57-101 -> LM:S:LI18:location:61-76 ->
LM:S:LI18:location:7-22 -> LM:S:LI18:location:84-101 -> RL:S:LI18:location:00:between -> RL:S:LI18:location:01:same-level ->
RL:S:LI18:location:02:surface-landmark -> S:LI18:location -> LI18

Final semantic execution result

```
{"location_statement_status":"RESOLVED","primary_location_status":"RESOLVED"}
```

Coordinates generated: 0 | Legacy C coordinate referenced: 0

```
[ ] semantic decomposition correct  
[ ] operator semantics correct  
[ ] operand identity correct  
[ ] dependency structure correct  
[ ] unresolved/conditional handling correct  
[ ] ACCEPT  
[ ] REJECT
```

LI17 - RESOLVED

WHO primary Location

S:LI17:location: On the anterior aspect of the neck, at the same level as the cricoid cartilage, just posterior to the border of the sternocleido-mastoid muscle.

Semantic decomposition / operands

LM:S:LI17:location:101-108
source: border
class: boundary.border | status: UNRESOLVED | executor: no_family_executor
identity: {"kind":null}
reason: unsupported landmark disposition/class: geometry_or_surface/boundary.border

LM:S:LI17:location:112-143
source: the sternocleido-mastoid muscle
class: muscle | status: RESOLVED | executor: fma_mesh
identity: {"fma_id":"FMA13407","kind":"fma_concept","part_ids":["FJ1573","FJ1595"]}
reason: None

LM:S:LI17:location:57-78
source: the cricoid cartilage
class: cartilage | status: RESOLVED | executor: fma_mesh
identity: {"fma_id":"FMA9615","kind":"fma_concept","part_ids":["FJ2440","FJ2769"]}
reason: None

LM:S:LI17:location:7-22
source: anterior aspect
class: surface.aspect | status: RESOLVED | executor: approved_surface_registry
identity: {"geometry_hash":"71f7a5ab627b3c93a9aceb574aabdb27ca3577688316b7de105a5effa7840dc7","kind":"surface_registry","registry_id":"SR:anterior_neck"}
reason: None

LM:S:LI17:location:97-143
source: the border of the sternocleido-mastoid muscle
class: muscle | status: RESOLVED | executor: fma_mesh
identity: {"fma_id":"FMA13407","kind":"fma_concept","part_ids":["FJ1573","FJ1595"]}
reason: None

LM:S:LI17:note:1:26-31
source: LI18
class: acupoint_reference | status: RESOLVED | executor: reference_acupoint
identity: {"kind":"reference_acupoint","reference_point_id":"LI18"}
reason: None

LM:S:LI17:note:1:54-58
source: St10
class: acupoint_reference | status: RESOLVED | executor: reference_acupoint
identity: {"kind":"reference_acupoint","reference_point_id":"ST10"}
reason: None

Relation / operator chain

RL:S:LI17:location:00:relative-to
status: RESOLVED | operator: directional_relation
operand IDs: LM:S:LI17:location:97-143
direction: posterior | source statement: S:LI17:location
reason: None

RL:S:LI17:location:01:same-level
status: RESOLVED | operator: same_level_plane
operand IDs: LM:S:LI17:location:57-78
direction: None | source statement: S:LI17:location
reason: None

RL:S:LI17:location:02:surface-landmark
status: RESOLVED | operator: surface_membership
operand IDs: LM:S:LI17:location:7-22
direction: None | source statement: S:LI17:location
reason: None

RL:S:LI17:note:1:00:relative-to
status: RESOLVED | operator: directional_relation
operand IDs: LM:S:LI17:note:1:26-31
direction: inferior | source statement: S:LI17:note:1
reason: None

RL:S:LI17:note:1:01:same-level
status: RESOLVED | operator: same_level_plane
operand IDs: LM:S:LI17:note:1:54-58
direction: None | source statement: S:LI17:note:1
reason: None

RL21:S:LI17:note:1:27-31:reference-acupoint:9ab9a5cce4
status: RESOLVED | operator: reference_dependency
operand IDs: LM:S:LI17:note:1:26-31
direction: None | source statement: S:LI17:note:1
reason: None

RL21:S:LI17:note:1:54-58:reference-acupoint:87e9101b1e
status: RESOLVED | operator: reference_dependency
operand IDs: LM:S:LI17:note:1:54-58
direction: None | source statement: S:LI17:note:1
reason: None

Dependency edges (projection of existing explicit IDs)

LM:S:LI17:location:57-78 -> RL:S:LI17:location:01:same-level [operand_to_relation]
LM:S:LI17:location:7-22 -> RL:S:LI17:location:02:surface-landmark [operand_to_relation]
LM:S:LI17:location:97-143 -> RL:S:LI17:location:00:relative-to [operand_to_relation]
LM:S:LI17:note:1:26-31 -> RL21:S:LI17:note:1:27-31:reference-acupoint:9ab9a5cce4 [operand_to_relation]
LM:S:LI17:note:1:26-31 -> RL:S:LI17:note:1:00:relative-to [operand_to_relation]
LM:S:LI17:note:1:54-58 -> RL21:S:LI17:note:1:54-58:reference-acupoint:87e9101b1e [operand_to_relation]
LM:S:LI17:note:1:54-58 -> RL:S:LI17:note:1:01:same-level [operand_to_relation]
RL21:S:LI17:note:1:27-31:reference-acupoint:9ab9a5cce4 -> S:LI17:note:1 [relation_to_statement]
RL21:S:LI17:note:1:54-58:reference-acupoint:87e9101b1e -> S:LI17:note:1 [relation_to_statement]
RL:S:LI17:location:00:relative-to -> S:LI17:location [relation_to_statement]
RL:S:LI17:location:01:same-level -> S:LI17:location [relation_to_statement]
RL:S:LI17:location:02:surface-landmark -> S:LI17:location [relation_to_statement]
RL:S:LI17:note:1:00:relative-to -> S:LI17:note:1 [relation_to_statement]
RL:S:LI17:note:1:01:same-level -> S:LI17:note:1 [relation_to_statement]
S:LI17:location -> LI17 [statement_to_point]
S:LI17:note:1 -> LI17 [statement_to_point]

Recorded data order

Stage 2 output does not record timestamped per-operation runtime order. The following preserves exact serialized Stage 2 data order.

LM:S:LI17:location:101-108 -> LM:S:LI17:location:112-143 -> LM:S:LI17:location:57-78 -> LM:S:LI17:location:7-22 -> LM:S:LI17:location:97-143 -> LM:S:LI17:note:1:26-31 -> LM:S:LI17:note:1:54-58 -> RL:S:LI17:location:00:relative-to -> RL:S:LI17:location:01:same-level -> RL:S:LI17:location:02:surface-landmark -> RL:S:LI17:note:1:00:relative-to -> RL:S:LI17:note:1:01:same-level -> RL21:S:LI17:note:1:27-31:reference-acupoint:9ab9a5cce4 -> RL21:S:LI17:note:1:54-58:reference-acupoint:87e9101b1e -> S:LI17:location -> S:LI17:note:1 -> LI17

Final semantic execution result

{"location_statement_status":"RESOLVED","primary_location_status":"RESOLVED"}

Coordinates generated: 0 | Legacy C coordinate referenced: 0

- [] semantic decomposition correct
- [] operator semantics correct
- [] operand identity correct
- [] dependency structure correct
- [] unresolved/conditional handling correct
- [] ACCEPT
- [] REJECT

BL17 - RESOLVED

WHO primary Location

S:BL17:location: In the upper back region, at the same level as the inferior border of the spinous process of the seventh thoracic vertebra (T7), 1.5 B-cun lateral to the posterior median line.

Semantic decomposition / operands

LM:S:BL17:location:149-175
source: the posterior median line
class: line.anatomical_line | status: RESOLVED | executor: approved_surface_registry
identity: {"geometry_hash":"62a4312f07e04de5ebb4791906ed7098bb7c85bea907953c59c8c92bec7d9f4d","kind":"surface_registry","registry_id":"SR:posterior_median_line"}
reason: None

LM:S:BL17:location:47-127
source: the inferior border of the spinous process of the seventh thoracic vertebra (T7)
class: bone.bone | status: RESOLVED | executor: fma_mesh
identity: {"fma_id":"FMA9968","kind":"fma_concept","part_ids":["FJ3173"]}
reason: None

LM:S:BL17:location:51-66
source: inferior border
class: boundary.border | status: RESOLVED | executor: composite_subfeature
identity: {"kind":"entity_subfeature","modifier":"inferior","parent_landmark_id":"LM:S:BL17:location:70-127"}
reason: None

LM:S:BL17:location:7-24
source: upper back region
class: region.body_region | status: RESOLVED | executor: approved_surface_registry
identity: {"geometry_hash":"86974891de98cccd457e268703bc6a672fce4777ca5d0776994671dd7f004323","kind":"surface_registry","registry_id":"SR:upper_back"}
reason: None

LM:S:BL17:location:70-127
source: the spinous process of the seventh thoracic vertebra (T7)
class: bone.bone | status: RESOLVED | executor: fma_mesh
identity: {"fma_id":"FMA9968","kind":"fma_concept","part_ids":["FJ3173"]}
reason: None

LM:S:BL17:note:1:64-116
source: the spinous process of the seventh thoracic vertebra
class: bone.bone | status: RESOLVED | executor: fma_mesh
identity: {"fma_id":"FMA9968","kind":"fma_concept","part_ids":["FJ3173"]}
reason: None

LM:S:BL17:note:1:75-83
source: process
class: bone.process_or_prominence | status: RESOLVED | executor: atlas_name_resolution
identity: {"kind":"atlas_part"}
reason: None

LM:S:BL17:note:1:87-116
source: the seventh thoracic vertebra
class: bone.bone | status: RESOLVED | executor: fma_mesh
identity: {"fma_id":"FMA9968","kind":"fma_concept","part_ids":["FJ3173"]}
reason: None

Relation / operator chain

RL:S:BL17:location:00:relative-to
status: RESOLVED | operator: directional_relation
operand IDs: LM:S:BL17:location:149-175
direction: lateral | source statement: S:BL17:location
reason: None

RL:S:BL17:location:01:same-level
status: RESOLVED | operator: same_level_plane
operand IDs: LM:S:BL17:location:47-127
direction: None | source statement: S:BL17:location
reason: None

RL:S:BL17:location:02:surface-landmark
status: RESOLVED | operator: surface_membership
operand IDs: LM:S:BL17:location:7-24
direction: None | source statement: S:BL17:location
reason: None

RL:S:BL17:note:1:00:same-level
status: RESOLVED | operator: same_level_plane
operand IDs: LM:S:BL17:note:1:64-116
direction: None | source statement: S:BL17:note:1
reason: None

Measurements / calibration use

{ "anchor_landmark_id": "LM:S:BL17:location:149-175", "direction": "lateral", "executor": "approved_calibration", "kind": "proportional_offset", "measurement_id": "PM:S:BL17:location:00", "reason": null, "source_statement_id": "S:BL17:location", "source_unit": "B-cun", "status": "RESOLVED", "unit": "B-cun", "value": 1.5, "who_pdf_page": 117 }

Dependency edges (projection of existing explicit IDs)

LM:S:BL17:location:149-175 -> PM:S:BL17:location:00 [anchor_to_measurement]
PM:S:BL17:location:00 -> S:BL17:location [measurement_to_statement]
LM:S:BL17:location:149-175 -> RL:S:BL17:location:00:relative-to [operand_to_relation]
LM:S:BL17:location:47-127 -> RL:S:BL17:location:01:same-level [operand_to_relation]
LM:S:BL17:location:7-24 -> RL:S:BL17:location:02:surface-landmark [operand_to_relation]
LM:S:BL17:note:1:64-116 -> RL:S:BL17:note:1:00:same-level [operand_to_relation]
RL:S:BL17:location:00:relative-to -> S:BL17:location [relation_to_statement]
RL:S:BL17:location:01:same-level -> S:BL17:location [relation_to_statement]
RL:S:BL17:location:02:surface-landmark -> S:BL17:location [relation_to_statement]
RL:S:BL17:note:1:00:same-level -> S:BL17:note:1 [relation_to_statement]
S:BL17:location -> BL17 [statement_to_point]
S:BL17:note:1 -> BL17 [statement_to_point]

Recorded data order

Stage 2 output does not record timestamped per-operation runtime order. The following preserves exact serialized Stage 2 data order.

LM:S:BL17:location:149-175 -> LM:S:BL17:location:47-127 -> LM:S:BL17:location:51-66 -> LM:S:BL17:location:7-24 -> LM:S:BL17:location:70-127 -> LM:S:BL17:note:1:64-116 -> LM:S:BL17:note:1:75-83 -> LM:S:BL17:note:1:87-116 -> PM:S:BL17:location:00 -> RL:S:BL17:location:00:relative-to -> RL:S:BL17:location:01:same-level -> RL:S:BL17:location:02:surface-landmark -> RL:S:BL17:note:1:00:same-level -> S:BL17:location -> S:BL17:note:1 -> BL17

Final semantic execution result

{ "location_statement_status": "RESOLVED", "primary_location_status": "RESOLVED" }

Coordinates generated: 0 | Legacy C coordinate referenced: 0

- [] semantic decomposition correct
- [] operator semantics correct
- [] operand identity correct
- [] dependency structure correct
- [] unresolved/conditional handling correct
- [] ACCEPT
- [] REJECT

BL23 - RESOLVED

WHO primary Location

S:BL23:location: In the lumbar region, at the same level as the inferior border of the spinous process of the second lumbar vertebra (L2), 1.5 B-cun lateral to the posterior median line.

Semantic decomposition / operands

LM:S:BL23:location:142-168
source: the posterior median line
class: line.anatomical_line | status: RESOLVED | executor: approved_surface_registry
identity: {"geometry_hash":"62a4312f07e04de5ebb4791906ed7098bb7c85bea907953c59c8c92bec7d9f4d","kind":"surface_registry","registry_id":"SR:posterior_median_line"}
reason: None

LM:S:BL23:location:43-120
source: the inferior border of the spinous process of the second lumbar vertebra (L2)
class: bone.bone | status: RESOLVED | executor: fma_mesh
identity: {"fma_id":"FMA13073","kind":"fma_concept","part_ids":["FJ3159"]}
reason: None

LM:S:BL23:location:47-62
source: inferior border
class: boundary.border | status: RESOLVED | executor: composite_subfeature
identity: {"kind":"entity_subfeature","modifier":"inferior","parent_landmark_id":"LM:S:BL23:location:66-120"}
reason: None

LM:S:BL23:location:66-120
source: the spinous process of the second lumbar vertebra (L2)
class: bone.bone | status: RESOLVED | executor: fma_mesh
identity: {"fma_id":"FMA13073","kind":"fma_concept","part_ids":["FJ3159"]}
reason: None

LM:S:BL23:location:7-20
source: lumbar region
class: region.body_region | status: RESOLVED | executor: approved_surface_registry
identity: {"geometry_hash":"ef7e2ce29e7bdfb2606e98c7c2254ff2a0c99337bcf65192272b78f367ce7244","kind":"surface_registry","registry_id":"SR:lumbar_region"}
reason: None

Relation / operator chain

RL:S:BL23:location:00:relative-to
status: RESOLVED | operator: directional_relation
operand IDs: LM:S:BL23:location:142-168
direction: lateral | source statement: S:BL23:location
reason: None

RL:S:BL23:location:01:same-level
status: RESOLVED | operator: same_level_plane
operand IDs: LM:S:BL23:location:43-120
direction: None | source statement: S:BL23:location
reason: None

RL:S:BL23:location:02:surface-landmark
status: RESOLVED | operator: surface_membership
operand IDs: LM:S:BL23:location:7-20
direction: None | source statement: S:BL23:location
reason: None

Measurements / calibration use

{"anchor_landmark_id":"LM:S:BL23:location:142-168","direction":"lateral","executor":"approved_calibration","kind":"proportional_offset","measurement_id":"PM:S:BL23:location:00","reason":null,"source_statement_id":"S:BL23:location","source_unit":"B-cun","status":"RESOLVED","unit":"B-cun","value":1.5,"who_pdf_page":120}

Dependency edges (projection of existing explicit IDs)

LM:S:BL23:location:142-168 -> PM:S:BL23:location:00 [anchor_to_measurement]
PM:S:BL23:location:00 -> S:BL23:location [measurement_to_statement]
LM:S:BL23:location:142-168 -> RL:S:BL23:location:00:relative-to [operand_to_relation]
LM:S:BL23:location:43-120 -> RL:S:BL23:location:01:same-level [operand_to_relation]
LM:S:BL23:location:7-20 -> RL:S:BL23:location:02:surface-landmark [operand_to_relation]
RL:S:BL23:location:00:relative-to -> S:BL23:location [relation_to_statement]
RL:S:BL23:location:01:same-level -> S:BL23:location [relation_to_statement]
RL:S:BL23:location:02:surface-landmark -> S:BL23:location [relation_to_statement]
S:BL23:location -> BL23 [statement_to_point]

Recorded data order

Stage 2 output does not record timestamped per-operation runtime order. The following preserves exact serialized Stage 2 data order.

LM:S:BL23:location:142-168 -> LM:S:BL23:location:43-120 -> LM:S:BL23:location:47-62 -> LM:S:BL23:location:66-120 -> LM:S:BL23:location:7-20 -> PM:S:BL23:location:00 -> RL:S:BL23:location:00:relative-to -> RL:S:BL23:location:01:same-level -> RL:S:BL23:location:02:surface-landmark -> S:BL23:location -> BL23

Final semantic execution result

```
{"location_statement_status":"RESOLVED","primary_location_status":"RESOLVED"}
```

Coordinates generated: 0 | Legacy C coordinate referenced: 0

- [] semantic decomposition correct
- [] operator semantics correct
- [] operand identity correct
- [] dependency structure correct
- [] unresolved/conditional handling correct
- [] ACCEPT
- [] REJECT

BL25 - RESOLVED

WHO primary Location

S:BL25:location: In the lumbar region, at the same level as the inferior border of the spinous process of the fourth lumbar vertebra (L4), 1.5 B-cun lateral to the posterior median line.

Semantic decomposition / operands

LM:S:BL25:location:142-168
source: the posterior median line
class: line.anatomical_line | status: RESOLVED | executor: approved_surface_registry
identity: {"geometry_hash":"62a4312f07e04de5ebb4791906ed7098bb7c85bea907953c59c8c92bec7d9f4d","kind":"surface_registry","registry_id":"SR:posterior_median_line"}
reason: None

LM:S:BL25:location:43-120
source: the inferior border of the spinous process of the fourth lumbar vertebra (L4)
class: bone.bone | status: RESOLVED | executor: fma_mesh
identity: {"fma_id":"FMA13075","kind":"fma_concept","part_ids":["FJ3165"]}
reason: None

LM:S:BL25:location:47-62
source: inferior border
class: boundary.border | status: RESOLVED | executor: composite_subfeature
identity: {"kind":"entity_subfeature","modifier":"inferior","parent_landmark_id":"LM:S:BL25:location:66-120"}
reason: None

LM:S:BL25:location:66-120
source: the spinous process of the fourth lumbar vertebra (L4)
class: bone.bone | status: RESOLVED | executor: fma_mesh
identity: {"fma_id":"FMA13075","kind":"fma_concept","part_ids":["FJ3165"]}
reason: None

LM:S:BL25:location:7-20
source: lumbar region
class: region.body_region | status: RESOLVED | executor: approved_surface_registry
identity: {"geometry_hash":"ef7e2ce29e7bdfb2606e98c7c2254ff2a0c99337bcf65192272b78f367ce7244","kind":"surface_registry","registry_id":"SR:lumbar_region"}
reason: None

Relation / operator chain

RL:S:BL25:location:00:relative-to
status: RESOLVED | operator: directional_relation
operand IDs: LM:S:BL25:location:142-168
direction: lateral | source statement: S:BL25:location
reason: None

RL:S:BL25:location:01:same-level
status: RESOLVED | operator: same_level_plane
operand IDs: LM:S:BL25:location:43-120
direction: None | source statement: S:BL25:location
reason: None

RL:S:BL25:location:02:surface-landmark
status: RESOLVED | operator: surface_membership
operand IDs: LM:S:BL25:location:7-20
direction: None | source statement: S:BL25:location
reason: None

Measurements / calibration use

{"anchor_landmark_id":"LM:S:BL25:location:142-168","direction":"lateral","executor":"approved_calibration","kind":"proportional_offset","measurement_id":"PM:S:BL25:location:00","reason":null,"source_statement_id":"S:BL25:location","source_unit":"B-cun","status":"RESOLVED","unit":"B-cun","value":1.5,"who_pdf_page":121}

Dependency edges (projection of existing explicit IDs)

LM:S:BL25:location:142-168 -> PM:S:BL25:location:00 [anchor_to_measurement]
PM:S:BL25:location:00 -> S:BL25:location [measurement_to_statement]
LM:S:BL25:location:142-168 -> RL:S:BL25:location:00:relative-to [operand_to_relation]
LM:S:BL25:location:43-120 -> RL:S:BL25:location:01:same-level [operand_to_relation]
LM:S:BL25:location:7-20 -> RL:S:BL25:location:02:surface-landmark [operand_to_relation]
RL:S:BL25:location:00:relative-to -> S:BL25:location [relation_to_statement]
RL:S:BL25:location:01:same-level -> S:BL25:location [relation_to_statement]
RL:S:BL25:location:02:surface-landmark -> S:BL25:location [relation_to_statement]
S:BL25:location -> BL25 [statement_to_point]

Recorded data order

Stage 2 output does not record timestamped per-operation runtime order. The following preserves exact serialized Stage 2 data order.

LM:S:BL25:location:142-168 -> LM:S:BL25:location:43-120 -> LM:S:BL25:location:47-62 -> LM:S:BL25:location:66-120 -> LM:S:BL25:location:7-20 -> PM:S:BL25:location:00 -> RL:S:BL25:location:00:relative-to -> RL:S:BL25:location:01:same-level -> RL:S:BL25:location:02:surface-landmark -> S:BL25:location -> BL25

Final semantic execution result

```
{"location_statement_status":"RESOLVED","primary_location_status":"RESOLVED"}
```

Coordinates generated: 0 | Legacy C coordinate referenced: 0

```
[ ] semantic decomposition correct
[ ] operator semantics correct
[ ] operand identity correct
[ ] dependency structure correct
[ ] unresolved/conditional handling correct
[ ] ACCEPT
[ ] REJECT
```

BL40 - RESOLVED

WHO primary Location

S:BL40:location: On the posterior aspect of the knee, at the midpoint of the popliteal crease.

Semantic decomposition / operands

```
LM:S:BL40:location:56-76
source: the popliteal crease
class: crease.skin_crease | status: RESOLVED | executor: approved_surface_registry
identity: {"geometry_hash":"1dbdd6b2b621a4ee0185cf93253f7062de8485746c285cadd13ae1177b60ff9","kind":"surface_registry","registry_id":"SR:popliteal_crease"}
reason: None

LM:S:BL40:location:7-23
source: posterior aspect
class: surface.aspect | status: RESOLVED | executor: approved_surface_registry
identity: {"geometry_hash":"12ffee9bff3884af77a892f4d06484125686f5b1fc7f181d5dc0a994051fc63e","kind":"surface_registry","registry_id":"SR:posterior_knee"}
reason: None
```

Relation / operator chain

```
RL:S:BL40:location:00:midpoint-of-entity
status: RESOLVED | operator: entity_midpoint
operand IDs: LM:S:BL40:location:56-76
direction: None | source statement: S:BL40:location
reason: None

RL:S:BL40:location:01:surface-landmark
status: RESOLVED | operator: surface_membership
operand IDs: LM:S:BL40:location:7-23
direction: None | source statement: S:BL40:location
reason: None
```

Dependency edges (projection of existing explicit IDs)

```
LM:S:BL40:location:56-76 -> RL:S:BL40:location:00:midpoint-of-entity [operand_to_relation]
LM:S:BL40:location:7-23 -> RL:S:BL40:location:01:surface-landmark [operand_to_relation]
RL:S:BL40:location:00:midpoint-of-entity -> S:BL40:location [relation_to_statement]
RL:S:BL40:location:01:surface-landmark -> S:BL40:location [relation_to_statement]
S:BL40:location -> BL40 [statement_to_point]
```

Recorded data order

Stage 2 output does not record timestamped per-operation runtime order. The following preserves exact serialized Stage 2 data order.

```
LM:S:BL40:location:56-76 -> LM:S:BL40:location:7-23 -> RL:S:BL40:location:00:midpoint-of-entity -> RL:S:BL40:location:01:surface-landmark -> S:BL40:location -> BL40
```

Final semantic execution result

```
{"location_statement_status":"RESOLVED","primary_location_status":"RESOLVED"}
```

Coordinates generated: 0 | Legacy C coordinate referenced: 0

- [] semantic decomposition correct
- [] operator semantics correct
- [] operand identity correct
- [] dependency structure correct
- [] unresolved/conditional handling correct
- [] ACCEPT
- [] REJECT

ST4 - RESOLVED

CONDITIONAL state present in Stage 2 output.

WHO primary Location

S:ST4:location: On the face, 0.4 f-cun lateral to the angle of the mouth.

Semantic decomposition / operands

LM:S:ST4:location:33-56
source: the angle of the mouth
class: orifice_or_cavity | status: RESOLVED | executor: fma_mesh
identity: {"fma_id":"FMA49184","kind":"fma_concept","part_ids":["FJ1252","FJ1253","FJ2741","FJ2748","FJ2753","FJ2760","FJ2761","FJ2762","FJ2766","FJ2768","FJ3289"]}
reason: None

LM:S:ST4:location:7-11
source: face
class: region.body_region | status: RESOLVED | executor: fma_mesh
identity: {"fma_id":"FMA24728","kind":"fma_concept","part_ids":["FJ1252","FJ1253","FJ1282","FJ1285","FJ1286","FJ1289","FJ1294","FJ1295","FJ1297","FJ1298","FJ1299","FJ1301","FJ1302","FJ1304","FJ1305","FJ1306","FJ1308","FJ1309","FJ1317","FJ1320","FJ1322","FJ1323","FJ1324","FJ1328","FJ1331","FJ1336","FJ1337","FJ1340","FJ1345","FJ1346","FJ1348","FJ1349","FJ1350","FJ1352","FJ1353","FJ1355","FJ1356","FJ1357","FJ1359","FJ1360","FJ1368","FJ1371","FJ1373","FJ1374","FJ1375","FJ1379","FJ1382","FJ2554","FJ2555","FJ2556","FJ2557","FJ2558","FJ2741","FJ2748","FJ2753","FJ2760","FJ2761","FJ2762","FJ2766","FJ2768","FJ2772","FJ3199","FJ3201","FJ3263","FJ3265","FJ3269","FJ3272","FJ3273","FJ3281","FJ3287","FJ3289","FJ3369","FJ3371","FJ3375","FJ3378","FJ3379","FJ3386","FJ3392","FJ3394","FJ3395"]}
reason: None

LM:S:ST4:note:1:16-39
source: the angle of the mouth
class: orifice_or_cavity | status: RESOLVED | executor: fma_mesh
identity: {"fma_id":"FMA49184","kind":"fma_concept","part_ids":["FJ1252","FJ1253","FJ2741","FJ2748","FJ2753","FJ2760","FJ2761","FJ2762","FJ2766","FJ2768","FJ3289"]}
reason: None

LM:S:ST4:note:1:69-86
source: nasolabial sulcus
class: soft_tissue_feature | status: UNRESOLVED | executor: approved_surface_registry
identity: {"kind":null}
reason: No approved BodyParts3D geometric definition; deliberately unresolved.

LM:S:ST4:note:1:97-134
source: continuation of the nasolabial sulcus
class: soft_tissue_feature | status: UNRESOLVED | executor: approved_surface_registry
identity: {"kind":null}
reason: No approved BodyParts3D geometric definition; deliberately unresolved.

Relation / operator chain

RL:S:ST4:location:00:relative-to
status: RESOLVED | operator: directional_relation
operand IDs: LM:S:ST4:location:33-56
direction: lateral | source statement: S:ST4:location
reason: None

RL:S:ST4:location:01:surface-landmark
status: RESOLVED | operator: surface_membership
operand IDs: LM:S:ST4:location:7-11
direction: None | source statement: S:ST4:location
reason: None

RL:S:ST4:note:1:00:relative-to
status: RESOLVED | operator: directional_relation
operand IDs: LM:S:ST4:note:1:16-39
direction: lateral | source statement: S:ST4:note:1
reason: None

RL:S:ST4:note:1:01:surface-landmark
status: UNRESOLVED | operator: surface_membership
operand IDs: LM:S:ST4:note:1:69-86
direction: None | source statement: None
reason: argument landmark not uniquely executable

RL:S:ST4:note:1:02:surface-landmark
status: UNRESOLVED | operator: surface_membership
operand IDs: LM:S:ST4:note:1:97-134
direction: None | source statement: None
reason: argument landmark not uniquely executable

Measurements / calibration use

{"anchor_landmark_id":"LM:S:ST4:location:33-56","direction":"lateral","executor":"approved_calibration","kind":"proportional_offset","measurement_id":"PM:S:ST4:location:00","reason":null,"source_statement_id":"S:ST4:location","source_unit":"F-cun","status":"RESOLVED","unit":"F-cun","value":0.4,"who_pdf_page":56}

Conditions

```
{ "branch_id": "branch_1", "condition_id": "CD:S:ST4:note:1:00", "condition_type": "alternative", "executor": "condition_contract", "reason": "branch preserved; not forced into default-pose primary computation", "source_span": { "char_end": 109, "char_start": 87, "condition_type": "alternative", "role": null, "source_raw": "or on the continuation" }, "source_statement_id": "S:ST4:note:1", "status": "CONDITIONAL" }
```

Dependency edges (projection of existing explicit IDs)

```
LM:S:ST4:location:33-56 -> PM:S:ST4:location:00 [anchor_to_measurement]
CD:S:ST4:note:1:00 -> S:ST4:note:1 [condition_to_statement]
PM:S:ST4:location:00 -> S:ST4:location [measurement_to_statement]
LM:S:ST4:location:33-56 -> RL:S:ST4:location:00:relative-to [operand_to_relation]
LM:S:ST4:location:7-11 -> RL:S:ST4:location:01:surface-landmark [operand_to_relation]
LM:S:ST4:note:1:16-39 -> RL:S:ST4:note:1:00:relative-to [operand_to_relation]
LM:S:ST4:note:1:69-86 -> RL:S:ST4:note:1:01:surface-landmark [operand_to_relation]
LM:S:ST4:note:1:97-134 -> RL:S:ST4:note:1:02:surface-landmark [operand_to_relation]
RL:S:ST4:location:00:relative-to -> S:ST4:location [relation_to_statement]
RL:S:ST4:location:01:surface-landmark -> S:ST4:location [relation_to_statement]
RL:S:ST4:note:1:00:relative-to -> S:ST4:note:1 [relation_to_statement]
S:ST4:location -> ST4 [statement_to_point]
S:ST4:note:1 -> ST4 [statement_to_point]
```

Recorded data order

Stage 2 output does not record timestamped per-operation runtime order. The following preserves exact serialized Stage 2 data order.

```
LM:S:ST4:location:33-56 -> LM:S:ST4:location:7-11 -> LM:S:ST4:note:1:16-39 -> LM:S:ST4:note:1:69-86 -> LM:S:ST4:note:1:97-134 ->
PM:S:ST4:location:00 -> CD:S:ST4:note:1:00 -> RL:S:ST4:location:00:relative-to -> RL:S:ST4:location:01:surface-landmark ->
RL:S:ST4:note:1:00:relative-to -> RL:S:ST4:note:1:01:surface-landmark -> RL:S:ST4:note:1:02:surface-landmark -> S:ST4:location -> S:ST4:note:1 ->
ST4
```

Final semantic execution result

```
{ "location_statement_status": "RESOLVED", "primary_location_status": "RESOLVED" }
```

Coordinates generated: 0 | Legacy C coordinate referenced: 0

- [] semantic decomposition correct
- [] operator semantics correct
- [] operand identity correct
- [] dependency structure correct
- [] unresolved/conditional handling correct
- [] ACCEPT
- [] REJECT

TE6 - RESOLVED

WHO primary Location

S:TE6:location: On the posterior aspect of the forearm, midpoint of the interosseous space between the radius and the ulna, 3 B-cun proximal to the dorsal wrist crease.

Semantic decomposition / operands

LM:S:TE6:location:127-151
source: the dorsal wrist crease
class: crease.skin_crease | status: RESOLVED | executor: approved_surface_registry
identity: {"geometry_hash":"0ff534739841ae12b463e9ea5102d7801e5de256871212f7ddeb84a0b0265288","kind":"surface_registry","registry_id":"SR:dorsal_wrist_crease"}
reason: None

LM:S:TE6:location:52-106
source: the interosseous space between the radius and the ulna
class: space.anatomical_space | status: RESOLVED | executor: approved_surface_registry
identity: {"geometry_hash":"4b89dd3e588ceb0c6fc9aae7de7f8c71ea842a9e0305158cad97d76f12f35f33","kind":"surface_registry","registry_id":"SR:radius_ulna_interosseous_space"}
reason: None

LM:S:TE6:location:7-23
source: posterior aspect
class: surface.aspect | status: RESOLVED | executor: approved_surface_registry
identity: {"geometry_hash":"7809ff04bb161d448c5b90dbbb407ddc76c10e74e85ee7fe0c795e866ebad73c","kind":"surface_registry","registry_id":"SR:posterior_forearm"}
reason: None

LM:S:TE6:location:87-93
source: radius
class: bone.bone | status: RESOLVED | executor: fma_mesh
identity: {"fma_id":"FMA23463","kind":"fma_concept","part_ids":["FJ3277","FJ3349"]}
reason: None

LM:S:TE6:location:98-106
source: the ulna
class: bone.bone | status: RESOLVED | executor: fma_mesh
identity: {"fma_id":"FMA23466","kind":"fma_concept","part_ids":["FJ3286","FJ3391"]}
reason: None

LM:S:TE6:note:1:25-29
source: TE5
class: acupoint_reference | status: RESOLVED | executor: reference_acupoint
identity: {"kind":"reference_acupoint","reference_point_id":"TE5"}
reason: None

LM:S:TE6:note:1:43-49
source: radius
class: bone.bone | status: RESOLVED | executor: fma_mesh
identity: {"fma_id":"FMA23463","kind":"fma_concept","part_ids":["FJ3277","FJ3349"]}
reason: None

LM:S:TE6:note:1:54-62
source: the ulna
class: bone.bone | status: RESOLVED | executor: fma_mesh
identity: {"fma_id":"FMA23466","kind":"fma_concept","part_ids":["FJ3286","FJ3391"]}
reason: None

LM:S:TE6:note:1:85-88
source: tE7
class: acupoint_reference | status: RESOLVED | executor: reference_acupoint
identity: {"kind":"reference_acupoint","reference_point_id":"TE7"}
reason: None

Relation / operator chain

RL:S:TE6:location:00:between
status: RESOLVED | operator: between_constraint
operand IDs: LM:S:TE6:location:87-93, LM:S:TE6:location:98-106
direction: None | source statement: S:TE6:location
reason: None

RL:S:TE6:location:01:relative-to
status: RESOLVED | operator: directional_relation
operand IDs: LM:S:TE6:location:127-151
direction: proximal | source statement: S:TE6:location
reason: None

RL:S:TE6:location:02:midpoint-of-entity
status: RESOLVED | operator: entity_midpoint
operand IDs: LM:S:TE6:location:52-106
direction: None | source statement: S:TE6:location
reason: None

RL:S:TE6:location:03:surface-landmark
status: RESOLVED | operator: surface_membership
operand IDs: LM:S:TE6:location:7-23
direction: None | source statement: S:TE6:location
reason: None

RL:S:TE6:note:1:00:between
status: RESOLVED | operator: between_constraint
operand IDs: LM:S:TE6:note:1:43-49, LM:S:TE6:note:1:54-62
direction: None | source statement: S:TE6:note:1
reason: None

RL:S:TE6:note:1:01:relative-to
status: RESOLVED | operator: directional_relation
operand IDs: LM:S:TE6:note:1:25-29
direction: proximal | source statement: S:TE6:note:1
reason: None

RL:S:TE6:note:1:02:same-level
status: RESOLVED | operator: same_level_plane
operand IDs: LM:S:TE6:note:1:85-88
direction: None | source statement: S:TE6:note:1
reason: None

RL21:S:TE6:note:1:26-29:reference-acupoint:87571bdea7
status: RESOLVED | operator: reference_dependency
operand IDs: LM:S:TE6:note:1:25-29
direction: None | source statement: S:TE6:note:1
reason: None

RL21:S:TE6:note:1:85-88:reference-acupoint:eeb3b40f1e
status: RESOLVED | operator: reference_dependency
operand IDs: LM:S:TE6:note:1:85-88
direction: None | source statement: S:TE6:note:1
reason: None

Measurements / calibration use

{"anchor_landmark_id":"LM:S:TE6:location:127-151","direction":"proximal","executor":"approved_calibration","kind":"proportional_offset","measurement_id":"PM:S:TE6:location:00","reason":null,"source_statement_id":"S:TE6:location","source_unit":"B-cun","status":"RESOLVED","unit":"B-cun","value":3.0,"who_pdf_page":169}

{"anchor_landmark_id":"LM:S:TE6:note:1:25-29","direction":"proximal","executor":"approved_calibration","kind":"proportional_offset","measurement_id":"PM:S:TE6:note:1:00","reason":null,"source_statement_id":"S:TE6:note:1","source_unit":"B-cun","status":"RESOLVED","unit":"B-cun","value":1.0,"who_pdf_page":169}

Dependency edges (projection of existing explicit IDs)

LM:S:TE6:location:127-151 -> PM:S:TE6:location:00 [anchor_to_measurement]

LM:S:TE6:note:1:25-29 -> PM:S:TE6:note:1:00 [anchor_to_measurement]

PM:S:TE6:location:00 -> S:TE6:location [measurement_to_statement]

PM:S:TE6:note:1:00 -> S:TE6:note:1 [measurement_to_statement]

LM:S:TE6:location:127-151 -> RL:S:TE6:location:01:relative-to [operand_to_relation]

LM:S:TE6:location:52-106 -> RL:S:TE6:location:02:midpoint-of-entity [operand_to_relation]

LM:S:TE6:location:7-23 -> RL:S:TE6:location:03:surface-landmark [operand_to_relation]

LM:S:TE6:location:87-93 -> RL:S:TE6:location:00:between [operand_to_relation]

LM:S:TE6:location:98-106 -> RL:S:TE6:location:00:between [operand_to_relation]

LM:S:TE6:note:1:25-29 -> RL21:S:TE6:note:1:26-29:reference-acupoint:87571bdea7 [operand_to_relation]

LM:S:TE6:note:1:25-29 -> RL:S:TE6:note:1:01:relative-to [operand_to_relation]

LM:S:TE6:note:1:43-49 -> RL:S:TE6:note:1:00:between [operand_to_relation]

LM:S:TE6:note:1:54-62 -> RL:S:TE6:note:1:00:between [operand_to_relation]

LM:S:TE6:note:1:85-88 -> RL21:S:TE6:note:1:85-88:reference-acupoint:eeb3b40f1e [operand_to_relation]

LM:S:TE6:note:1:85-88 -> RL:S:TE6:note:1:02:same-level [operand_to_relation]

RL21:S:TE6:note:1:26-29:reference-acupoint:87571bdea7 -> S:TE6:note:1 [relation_to_statement]

RL21:S:TE6:note:1:85-88:reference-acupoint:eeb3b40f1e -> S:TE6:note:1 [relation_to_statement]

RL:S:TE6:location:00:between -> S:TE6:location [relation_to_statement]

RL:S:TE6:location:01:relative-to -> S:TE6:location [relation_to_statement]

RL:S:TE6:location:02:midpoint-of-entity -> S:TE6:location [relation_to_statement]

RL:S:TE6:location:03:surface-landmark -> S:TE6:location [relation_to_statement]
RL:S:TE6:note:1:00:between -> S:TE6:note:1 [relation_to_statement]
RL:S:TE6:note:1:01:relative-to -> S:TE6:note:1 [relation_to_statement]
RL:S:TE6:note:1:02:same-level -> S:TE6:note:1 [relation_to_statement]
S:TE6:location -> TE6 [statement_to_point]
S:TE6:note:1 -> TE6 [statement_to_point]

Recorded data order

Stage 2 output does not record timestamped per-operation runtime order. The following preserves exact serialized Stage 2 data order.

LM:S:TE6:location:127-151 -> LM:S:TE6:location:52-106 -> LM:S:TE6:location:7-23 -> LM:S:TE6:location:87-93 -> LM:S:TE6:location:98-106 -> LM:S:TE6:note:1:25-29 -> LM:S:TE6:note:1:43-49 -> LM:S:TE6:note:1:54-62 -> LM:S:TE6:note:1:85-88 -> PM:S:TE6:location:00 -> PM:S:TE6:note:1:00 -> RL:S:TE6:location:00:between -> RL:S:TE6:location:01:relative-to -> RL:S:TE6:location:02:midpoint-of-entity -> RL:S:TE6:location:03:surface-landmark -> RL:S:TE6:note:1:00:between -> RL:S:TE6:note:1:01:relative-to -> RL:S:TE6:note:1:02:same-level -> RL21:S:TE6:note:1:26-29:reference-acupoint:87571bdea7 -> RL21:S:TE6:note:1:85-88:reference-acupoint:eeb3b40f1e -> S:TE6:location -> S:TE6:note:1 -> TE6

Final semantic execution result

{"location_statement_status":"RESOLVED","primary_location_status":"RESOLVED"}

Coordinates generated: 0 | Legacy C coordinate referenced: 0

[] semantic decomposition correct
[] operator semantics correct
[] operand identity correct
[] dependency structure correct
[] unresolved/conditional handling correct
[] ACCEPT
[] REJECT

ST9 - RESOLVED

CONDITIONAL state present in Stage 2 output.

WHO primary Location

S:ST9:location: In the anterior region of the neck, at the same level as the superior border of the thyroid cartilage, anterior to the sterno-cleidomastoid muscle, over the common carotid artery.

Semantic decomposition / operands

LM:S:ST9:location:114-146
source: the sterno-cleidomastoid muscle
class: muscle | status: RESOLVED | executor: fma_mesh
identity: {"fma_id":"FMA13407","kind":"fma_concept","part_ids":["FJ1573","FJ1595"]}
reason: None

LM:S:ST9:location:157-178
source: common carotid artery
class: vessel | status: RESOLVED | executor: fma_mesh
identity: {"fma_id":"FMA3939","kind":"fma_concept","part_ids":["FJ3483","FJ3564"]}
reason: None

LM:S:ST9:location:57-101
source: the superior border of the thyroid cartilage
class: cartilage | status: RESOLVED | executor: fma_mesh
identity: {"fma_id":"FMA55099","kind":"fma_concept","part_ids":["FJ2808"]}
reason: None

LM:S:ST9:location:61-76
source: superior border
class: boundary.border | status: RESOLVED | executor: composite_subfeature
identity: {"kind":"entity_subfeature","modifier":"superior","parent_landmark_id":"LM:S:ST9:location:84-101"}
reason: None

LM:S:ST9:location:7-34
source: anterior region of the neck
class: region.body_region | status: RESOLVED | executor: approved_surface_registry
identity: {"geometry_hash":"71f7a5ab627b3c93a9aceb574aabdb27ca3577688316b7de105a5effa7840dc7","kind":"surface_registry","registry_id":"SR:anterior_neck"}
reason: None

LM:S:ST9:location:84-101
source: thyroid cartilage
class: cartilage | status: RESOLVED | executor: fma_mesh
identity: {"fma_id":"FMA55099","kind":"fma_concept","part_ids":["FJ2808"]}
reason: None

LM:S:ST9:note:2:131-162
source: the sternocleidomastoid muscle
class: muscle | status: RESOLVED | executor: fma_mesh
identity: {"fma_id":"FMA13407","kind":"fma_concept","part_ids":["FJ1573","FJ1595"]}
reason: None

LM:S:ST9:note:2:186-226
source: the sternocleidomastoid muscle and LI18
class: muscle | status: RESOLVED | executor: fma_mesh
identity: {"fma_id":"FMA13407","kind":"fma_concept","part_ids":["FJ1573","FJ1595"]}
reason: None

LM:S:ST9:note:2:240-305:6f7a364e
source: anterior and pos-terior borders of the sternocleidomastoid muscle
class: muscle | status: RESOLVED | executor: fma_mesh
identity: {"fma_id":"FMA13407","kind":"fma_concept","part_ids":["FJ1573","FJ1595"]}
reason: None

LM:S:ST9:note:2:253-305:a442a1a4
source: pos-terior borders of the sternocleidomastoid muscle
class: muscle | status: RESOLVED | executor: fma_mesh
identity: {"fma_id":"FMA13407","kind":"fma_concept","part_ids":["FJ1573","FJ1595"]}
reason: None

LM:S:ST9:note:2:44-48
source: LI18
class: acupoint_reference | status: RESOLVED | executor: reference_acupoint
identity: {"kind":"reference_acupoint","reference_point_id":"LI18"}
reason: None

LM:S:ST9:note:2:63-78
source: superior border
class: boundary.border | status: RESOLVED | executor: composite_subfeature
identity: {"kind":"entity_subfeature","modifier":"superior","parent_landmark_id":"LM:S:ST9:note:2:82-103"}
reason: None

LM:S:ST9:note:2:82-103
source: the thyroid cartilage
class: cartilage | status: RESOLVED | executor: fma_mesh
identity: {"fma_id":"FMA55099","kind":"fma_concept","part_ids":["FJ2808"]}
reason: None

LM21:S:ST9:note:2:50-54:6775aaafeb
source: SI16
class: acupoint_reference | status: RESOLVED | executor: reference_acupoint
identity: {"kind":"reference_acupoint","reference_point_id":"SI16"}
reason: None

LM21:S:ST9:note:2:168-172:6775aaafeb
source: SI16
class: acupoint_reference | status: RESOLVED | executor: reference_acupoint
identity: {"kind":"reference_acupoint","reference_point_id":"SI16"}
reason: None

LM21:S:ST9:note:2:222-226:34574dfae8
source: LI18
class: acupoint_reference | status: RESOLVED | executor: reference_acupoint
identity: {"kind":"reference_acupoint","reference_point_id":"LI18"}
reason: None

Relation / operator chain

RL:S:ST9:location:00:relative-to
status: RESOLVED | operator: directional_relation
operand IDs: LM:S:ST9:location:114-146
direction: anterior | source statement: S:ST9:location
reason: None

RL:S:ST9:location:01:same-level
status: RESOLVED | operator: same_level_plane
operand IDs: LM:S:ST9:location:57-101
direction: None | source statement: S:ST9:location
reason: None

RL:S:ST9:location:02:overlies
status: RESOLVED | operator: overlies_projection
operand IDs: LM:S:ST9:location:157-178
direction: None | source statement: S:ST9:location
reason: None

RL:S:ST9:location:03:surface-landmark
status: RESOLVED | operator: surface_membership
operand IDs: LM:S:ST9:location:7-34
direction: None | source statement: S:ST9:location
reason: None

RL:S:ST9:note:2:00:between
status: RESOLVED | operator: between_constraint
operand IDs: LM:S:ST9:note:2:240-305:6f7a364e, LM:S:ST9:note:2:253-305:a442a1a4
direction: None | source statement: S:ST9:note:2
reason: None

RL:S:ST9:note:2:01:relative-to
status: RESOLVED | operator: directional_relation
operand IDs: LM:S:ST9:note:2:131-162
direction: anterior | source statement: S:ST9:note:2
reason: None

RL:S:ST9:note:2:02:relative-to
status: RESOLVED | operator: directional_relation
operand IDs: LM:S:ST9:note:2:186-226
direction: posterior | source statement: S:ST9:note:2
reason: None

RL:S:ST9:note:2:03:same-level
status: RESOLVED | operator: same_level_plane
operand IDs: LM:S:ST9:note:2:44-48
direction: None | source statement: S:ST9:note:2
reason: None

RL21:S:ST9:note:2:44-48:reference-acupoint:dfa4bc85d2
status: RESOLVED | operator: reference_dependency
operand IDs: LM:S:ST9:note:2:44-48
direction: None | source statement: S:ST9:note:2
reason: None

RL21:S:ST9:note:2:50-54:reference-acupoint:cf7c9e057c
status: RESOLVED | operator: reference_dependency
operand IDs: LM21:S:ST9:note:2:50-54:6775aaafeb
direction: None | source statement: S:ST9:note:2
reason: None

RL21:S:ST9:note:2:168-172:reference-acupoint:a3b2f0d073
status: RESOLVED | operator: reference_dependency
operand IDs: LM21:S:ST9:note:2:168-172:6775aaafeb
direction: None | source statement: S:ST9:note:2
reason: None

RL21:S:ST9:note:2:222-226:reference-acupoint:01273086b9
status: RESOLVED | operator: reference_dependency
operand IDs: LM21:S:ST9:note:2:222-226:34574dfae8
direction: None | source statement: S:ST9:note:2
reason: None

Conditions

{"branch_id":"branch_1","condition_id":"CD:S:ST9:note:1:00","condition_type":"body_position","executor":"condition_contract","reason":"condition requires pose incompatible with default BodyParts3D pose","source_span":{"char_end":119,"char_start":56,"condition_type":"body_position","role":null,"source_raw":"when the head is turned to the opposite side against resistance"},"source_statement_id":"S:ST9:note:1","status":"CONDITIONAL"}

{"branch_id":"branch_2","condition_id":"CD:S:ST9:note:1:01","condition_type":"palpation_dependent","executor":"condition_contract","reason":"branch preserved; not forced into default-pose primary computation","source_span":{"char_end":119,"char_start":42,"condition_type":"palpation_dependent","role":null,"source_raw":"more distinct when the head is turned to the opposite side against resistance"},"source_statement_id":"S:ST9:note:1","status":"CONDITIONAL"}

Dependency edges (projection of existing explicit IDs)

CD:S:ST9:note:1:00 -> S:ST9:note:1 [condition_to_statement]

CD:S:ST9:note:1:01 -> S:ST9:note:1 [condition_to_statement]

LM21:S:ST9:note:2:168-172:6775aaafeb -> RL21:S:ST9:note:2:168-172:reference-acupoint:a3b2f0d073 [operand_to_relation]

LM21:S:ST9:note:2:222-226:34574dfae8 -> RL21:S:ST9:note:2:222-226:reference-acupoint:01273086b9 [operand_to_relation]

LM21:S:ST9:note:2:50-54:6775aaafeb -> RL21:S:ST9:note:2:50-54:reference-acupoint:cf7c9e057c [operand_to_relation]

LM:S:ST9:location:114-146 -> RL:S:ST9:location:00:relative-to [operand_to_relation]

LM:S:ST9:location:157-178 -> RL:S:ST9:location:02:overlies [operand_to_relation]

LM:S:ST9:location:57-101 -> RL:S:ST9:location:01:same-level [operand_to_relation]

LM:S:ST9:location:7-34 -> RL:S:ST9:location:03:surface-landmark [operand_to_relation]

LM:S:ST9:note:2:131-162 -> RL:S:ST9:note:2:01:relative-to [operand_to_relation]

LM:S:ST9:note:2:186-226 -> RL:S:ST9:note:2:02:relative-to [operand_to_relation]

LM:S:ST9:note:2:240-305:6f7a364e -> RL:S:ST9:note:2:00:between [operand_to_relation]

LM:S:ST9:note:2:253-305:a442a1a4 -> RL:S:ST9:note:2:00:between [operand_to_relation]

LM:S:ST9:note:2:44-48 -> RL21:S:ST9:note:2:44-48:reference-acupoint:dfa4bc85d2 [operand_to_relation]

LM:S:ST9:note:2:44-48 -> RL:S:ST9:note:2:03:same-level [operand_to_relation]

RL21:S:ST9:note:2:168-172:reference-acupoint:a3b2f0d073 -> S:ST9:note:2 [relation_to_statement]

RL21:S:ST9:note:2:222-226:reference-acupoint:01273086b9 -> S:ST9:note:2 [relation_to_statement]

RL21:S:ST9:note:2:44-48:reference-acupoint:dfa4bc85d2 -> S:ST9:note:2 [relation_to_statement]

RL21:S:ST9:note:2:50-54:reference-acupoint:cf7c9e057c -> S:ST9:note:2 [relation_to_statement]

RL:S:ST9:location:00:relative-to -> S:ST9:location [relation_to_statement]

RL:S:ST9:location:01:same-level -> S:ST9:location [relation_to_statement]

RL:S:ST9:location:02:overlies -> S:ST9:location [relation_to_statement]

RL:S:ST9:location:03:surface-landmark -> S:ST9:location [relation_to_statement]

RL:S:ST9:note:2:00:between -> S:ST9:note:2 [relation_to_statement]

RL:S:ST9:note:2:01:relative-to -> S:ST9:note:2 [relation_to_statement]

RL:S:ST9:note:2:02:relative-to -> S:ST9:note:2 [relation_to_statement]

RL:S:ST9:note:2:03:same-level -> S:ST9:note:2 [relation_to_statement]

S:ST9:location -> ST9 [statement_to_point]

S:ST9:note:1 -> ST9 [statement_to_point]

S:ST9:note:2 -> ST9 [statement_to_point]

Recorded data order

Stage 2 output does not record timestamped per-operation runtime order. The following preserves exact serialized Stage 2 data order.

LM:S:ST9:location:114-146 -> LM:S:ST9:location:157-178 -> LM:S:ST9:location:57-101 -> LM:S:ST9:location:61-76 -> LM:S:ST9:location:7-34 ->
LM:S:ST9:location:84-101 -> LM:S:ST9:note:2:131-162 -> LM:S:ST9:note:2:186-226 -> LM:S:ST9:note:2:240-305:6f7a364e ->
LM:S:ST9:note:2:253-305:a442a1a4 -> LM:S:ST9:note:2:44-48 -> LM:S:ST9:note:2:63-78 -> LM:S:ST9:note:2:82-103 ->
LM21:S:ST9:note:2:50-54:6775aaafeb -> LM21:S:ST9:note:2:168-172:6775aaafeb -> LM21:S:ST9:note:2:222-226:34574dfae8 -> CD:S:ST9:note:1:00
-> CD:S:ST9:note:1:01 -> RL:S:ST9:location:00:relative-to -> RL:S:ST9:location:01:same-level -> RL:S:ST9:location:02:overlies ->
RL:S:ST9:location:03:surface-landmark -> RL:S:ST9:note:2:00:between -> RL:S:ST9:note:2:01:relative-to -> RL:S:ST9:note:2:02:relative-to ->
RL:S:ST9:note:2:03:same-level -> RL21:S:ST9:note:2:44-48:reference-acupoint:dfa4bc85d2 ->
RL21:S:ST9:note:2:50-54:reference-acupoint:cf7c9e057c -> RL21:S:ST9:note:2:168-172:reference-acupoint:a3b2f0d073 ->
RL21:S:ST9:note:2:222-226:reference-acupoint:01273086b9 -> S:ST9:location -> S:ST9:note:1 -> S:ST9:note:2 -> ST9

Final semantic execution result

{"location_statement_status":"RESOLVED","primary_location_status":"RESOLVED"}

Coordinates generated: 0 | Legacy C coordinate referenced: 0

[] semantic decomposition correct
[] operator semantics correct
[] operand identity correct
[] dependency structure correct
[] unresolved/conditional handling correct
[] ACCEPT
[] REJECT

HT7 - UNRESOLVED

WHO primary Location

S:HT7:location: On the anteromedial aspect of the wrist, radial to the flexor carpi ulnaris tendon, on the palmar wrist crease.

Semantic decomposition / operands

LM:S:HT7:location:50-82
source: the flexor carpi ulnaris tendon
class: tendon | status: UNRESOLVED | executor: atlas_name_resolution
identity: {"kind":null}
reason: no unique atlas name match

LM:S:HT7:location:7-26
source: anteromedial aspect
class: surface.aspect | status: RESOLVED | executor: approved_surface_registry
identity: {"geometry_hash":"6621b519a3e9551fb3372da3e363c6dd82762c2bc01e5e66f461ac31124c6054","kind":"surface_registry","registry_id":"SR:anteromedial_wrist"}
reason: None

LM:S:HT7:location:91-110
source: palmar wrist crease
class: crease.skin_crease | status: RESOLVED | executor: approved_surface_registry
identity: {"geometry_hash":"383e200f286d70d3c13689830673e15bb470bf299882294478ffe17ce465fa2f","kind":"surface_registry","registry_id":"SR:palmar_wrist_crease"}
reason: None

LM:S:HT7:note:1:13-23
source: depression
class: fossa_or_depression | status: UNRESOLVED | executor: explicit_unresolved_taxonomy
identity: {"kind":null}
reason: ambiguous_generic

LM:S:HT7:note:1:33-75
source: the proxi-mal border of the pisiform bone
class: boundary.border | status: UNRESOLVED | executor: no_family_executor
identity: {"kind":null}
reason: unsupported landmark disposition/class: geometry_or_surface/boundary.border

LM:S:HT7:note:1:38-54
source: proxi-mal border
class: boundary.border | status: UNRESOLVED | executor: no_family_executor
identity: {"kind":null}
reason: unsupported landmark disposition/class: geometry_or_surface/boundary.border

LM:S:HT7:note:1:58-75
source: the pisiform bone
class: bone.bone | status: MULTIPLE | executor: atlas_name_resolution
identity: {"candidates":[{"name":"Left pisiform","part_id":"FJ3276"}, {"name":"Right pisiform","part_id":"FJ3382"}], "kind":null}
reason: multiple atlas-name candidates

LM:S:HT7:note:1:84-103
source: palmar wrist crease
class: crease.skin_crease | status: RESOLVED | executor: approved_surface_registry
identity: {"geometry_hash":"383e200f286d70d3c13689830673e15bb470bf299882294478ffe17ce465fa2f","kind":"surface_registry","registry_id":"SR:palmar_wrist_crease"}
reason: None

Relation / operator chain

RL:S:HT7:location:00:relative-to
status: UNRESOLVED | operator: directional_relation
operand IDs: LM:S:HT7:location:50-82
direction: None | source statement: None
reason: argument landmark not uniquely executable

RL:S:HT7:location:01:surface-landmark
status: RESOLVED | operator: surface_membership
operand IDs: LM:S:HT7:location:7-26
direction: None | source statement: S:HT7:location
reason: None

RL:S:HT7:location:02:surface-landmark
status: RESOLVED | operator: surface_membership
operand IDs: LM:S:HT7:location:91-110
direction: None | source statement: S:HT7:location
reason: None

RL:S:HT7:note:1:00:relative-to
status: UNRESOLVED | operator: directional_relation
operand IDs: LM:S:HT7:note:1:33-75
direction: None | source statement: None
reason: argument landmark not uniquely executable

RL:S:HT7:note:1:01:surface-landmark
status: UNRESOLVED | operator: surface_membership
operand IDs: LM:S:HT7:note:1:13-23
direction: None | source statement: None
reason: argument landmark not uniquely executable

RL:S:HT7:note:1:03:surface-landmark
status: RESOLVED | operator: surface_membership
operand IDs: LM:S:HT7:note:1:84-103
direction: None | source statement: S:HT7:note:1
reason: None

Dependency edges (projection of existing explicit IDs)

LM:S:HT7:location:50-82 -> RL:S:HT7:location:00:relative-to [operand_to_relation]
LM:S:HT7:location:7-26 -> RL:S:HT7:location:01:surface-landmark [operand_to_relation]
LM:S:HT7:location:91-110 -> RL:S:HT7:location:02:surface-landmark [operand_to_relation]
LM:S:HT7:note:1:13-23 -> RL:S:HT7:note:1:01:surface-landmark [operand_to_relation]
LM:S:HT7:note:1:33-75 -> RL:S:HT7:note:1:00:relative-to [operand_to_relation]
LM:S:HT7:note:1:84-103 -> RL:S:HT7:note:1:03:surface-landmark [operand_to_relation]
RL:S:HT7:location:01:surface-landmark -> S:HT7:location [relation_to_statement]
RL:S:HT7:location:02:surface-landmark -> S:HT7:location [relation_to_statement]
RL:S:HT7:note:1:03:surface-landmark -> S:HT7:note:1 [relation_to_statement]
S:HT7:location -> HT7 [statement_to_point]
S:HT7:note:1 -> HT7 [statement_to_point]

Recorded data order

Stage 2 output does not record timestamped per-operation runtime order. The following preserves exact serialized Stage 2 data order.

LM:S:HT7:location:50-82 -> LM:S:HT7:location:7-26 -> LM:S:HT7:location:91-110 -> LM:S:HT7:note:1:13-23 -> LM:S:HT7:note:1:33-75 ->
LM:S:HT7:note:1:38-54 -> LM:S:HT7:note:1:58-75 -> LM:S:HT7:note:1:84-103 -> RL:S:HT7:location:00:relative-to ->
RL:S:HT7:location:01:surface-landmark -> RL:S:HT7:location:02:surface-landmark -> RL:S:HT7:note:1:00:relative-to ->
RL:S:HT7:note:1:01:surface-landmark -> RL:S:HT7:note:1:03:surface-landmark -> S:HT7:location -> S:HT7:note:1 -> HT7

Final semantic execution result

{ "location_statement_status": "UNRESOLVED", "primary_location_status": "UNRESOLVED" }

Blocking dependency

{ "blocking_items": [{ "landmark_class": "tendon", "node_id": "LM:S:HT7:location:50-82", "node_type": "landmark", "reason": "no unique atlas name match", "source_raw": " the flexor carpi ulnaris tendon", "status": "UNRESOLVED" }, { "downstream_stop_items": ["RL:S:HT7:location:00:relative-to"], "first_unresolved_dependency": { "landmark_class": "tendon", "node_id": "LM:S:HT7:location:50-82", "node_type": "landmark", "reason": "no unique atlas name match", "source_raw": " the flexor carpi ulnaris tendon", "status": "UNRESOLVED" }, "upstream_resolved_dependencies": ["RL:S:HT7:location:01:surface-landmark", "LM:S:HT7:location:7-26", "RL:S:HT7:location:02:surface-landmark", "LM:S:HT7:location:91-110"] }] }

Coordinates generated: 0 | Legacy C coordinate referenced: 0

- [] semantic decomposition correct
- [] operator semantics correct
- [] operand identity correct
- [] dependency structure correct
- [] unresolved/conditional handling correct
- [] ACCEPT
- [] REJECT

ST1 - UNRESOLVED

WHO primary Location

S:ST1:location: On the face, between the eyeball and the infraorbital margin, directly inferior to the pupil.

Semantic decomposition / operands

LM:S:ST1:location:25-32
source: eyeball
class: soft_tissue_feature | status: MULTIPLE | executor: atlas_name_resolution
identity: {"candidates":[{"name":"Anterior chamber of left eyeball","part_id":"FJ1282"},{"name":"Anterior chamber of right eyeball","part_id":"FJ1332"}],"kind":null}
reason: multiple atlas-name candidates

LM:S:ST1:location:37-60
source: the infraorbital margin
class: boundary.border | status: UNRESOLVED | executor: atlas_name_resolution
identity: {"kind":null}
reason: no unique atlas name match

LM:S:ST1:location:7-11
source: face
class: region.body_region | status: RESOLVED | executor: fma_mesh
identity: {"fma_id":"FMA24728","kind":"fma_concept","part_ids":["FJ1252","FJ1253","FJ1282","FJ1285","FJ1286","FJ1289","FJ1294","FJ1295","FJ1297","FJ1298","FJ1299","FJ1301","FJ1302","FJ1304","FJ1305","FJ1306","FJ1308","FJ1309","FJ1317","FJ1320","FJ1322","FJ1323","FJ1324","FJ1328","FJ1331","FJ1336","FJ1337","FJ1340","FJ1345","FJ1346","FJ1348","FJ1349","FJ1350","FJ1352","FJ1353","FJ1355","FJ1356","FJ1357","FJ1359","FJ1360","FJ1368","FJ1371","FJ1373","FJ1374","FJ1375","FJ1379","FJ1382","FJ2554","FJ2555","FJ2556","FJ2557","FJ2558","FJ2741","FJ2748","FJ2753","FJ2760","FJ2761","FJ2762","FJ2766","FJ2768","FJ2772","FJ3199","FJ3201","FJ3263","FJ3265","FJ3269","FJ3272","FJ3273","FJ3281","FJ3287","FJ3289","FJ3369","FJ3371","FJ3375","FJ3378","FJ3379","FJ3386","FJ3392","FJ3394","FJ3395"]}
reason: None

LM:S:ST1:location:82-92
source: the pupil
class: soft_tissue_feature | status: UNRESOLVED | executor: atlas_name_resolution
identity: {"kind":null}
reason: no unique atlas name match

Relation / operator chain

RL:S:ST1:location:00:between
status: UNRESOLVED | operator: between_constraint
operand IDs: LM:S:ST1:location:25-32, LM:S:ST1:location:37-60
direction: None | source statement: None
reason: argument landmark not uniquely executable

RL:S:ST1:location:01:relative-to
status: UNRESOLVED | operator: directional_relation
operand IDs: LM:S:ST1:location:82-92
direction: None | source statement: None
reason: argument landmark not uniquely executable

RL:S:ST1:location:02:surface-landmark
status: RESOLVED | operator: surface_membership
operand IDs: LM:S:ST1:location:7-11
direction: None | source statement: S:ST1:location
reason: None

Dependency edges (projection of existing explicit IDs)

LM:S:ST1:location:25-32 -> RL:S:ST1:location:00:between [operand_to_relation]

LM:S:ST1:location:37-60 -> RL:S:ST1:location:00:between [operand_to_relation]

LM:S:ST1:location:7-11 -> RL:S:ST1:location:02:surface-landmark [operand_to_relation]

LM:S:ST1:location:82-92 -> RL:S:ST1:location:01:relative-to [operand_to_relation]

RL:S:ST1:location:02:surface-landmark -> S:ST1:location [relation_to_statement]

S:ST1:location -> ST1 [statement_to_point]

Recorded data order

Stage 2 output does not record timestamped per-operation runtime order. The following preserves exact serialized Stage 2 data order.

LM:S:ST1:location:25-32 -> LM:S:ST1:location:37-60 -> LM:S:ST1:location:7-11 -> LM:S:ST1:location:82-92 -> RL:S:ST1:location:00:between -> RL:S:ST1:location:01:relative-to -> RL:S:ST1:location:02:surface-landmark -> S:ST1:location -> ST1

Final semantic execution result

{"location_statement_status":"UNRESOLVED","primary_location_status":"UNRESOLVED"}

Blocking dependency

```
{
  "blocking_items": [
    {
      "landmark_class": "soft_tissue_feature",
      "node_id": "LM:S:ST1:location:25-32",
      "node_type": "landmark",
      "reason": "multiple atlas-name candidates",
      "source_raw": "eyeball",
      "status": "MULTIPLE"
    },
    {
      "landmark_class": "boundary.border",
      "node_id": "LM:S:ST1:location:37-60",
      "node_type": "landmark",
      "reason": "no unique atlas name match",
      "source_raw": "the infraorbital margin",
      "status": "UNRESOLVED"
    },
    {
      "landmark_class": "soft_tissue_feature",
      "node_id": "LM:S:ST1:location:82-92",
      "node_type": "landmark",
      "reason": "no unique atlas name match",
      "source_raw": " the pupil",
      "status": "UNRESOLVED"
    }
  ],
  "downstream_stop_items": [
    {
      "RL:S:ST1:location:00:between",
      "RL:S:ST1:location:01:relative-to"
    }
  ],
  "first_unresolved_dependency": {
    "landmark_class": "soft_tissue_feature",
    "node_id": "LM:S:ST1:location:25-32",
    "node_type": "landmark",
    "reason": "multiple atlas-name candidates",
    "source_raw": "eyeball",
    "status": "MULTIPLE"
  },
  "upstream_resolved_dependencies": [
    {
      "RL:S:ST1:location:02:surface-landmark",
      "LM:S:ST1:location:7-11"
    }
  ]
}
```

Coordinates generated: 0 | Legacy C coordinate referenced: 0

- [] semantic decomposition correct
- [] operator semantics correct
- [] operand identity correct
- [] dependency structure correct
- [] unresolved/conditional handling correct
- [] ACCEPT
- [] REJECT

GB14 - UNRESOLVED

WHO primary Location

S:GB14:location: On the head, 1 B-cun superior to the eye-brow, directly superior to the centre of the pupil.

Semantic decomposition / operands

LM:S:GB14:location:32-45
source: the eye-brow
class: soft_tissue_feature | status: RESOLVED | executor: fma_mesh
identity: {"fma_id":"FMA54237","kind":"fma_concept","part_ids":["FJ2812"]}
reason: None

LM:S:GB14:location:67-91
source: the centre of the pupil
class: soft_tissue_feature | status: UNRESOLVED | executor: atlas_name_resolution
identity: {"kind":null}
reason: no unique atlas name match

LM:S:GB14:location:7-11
source: head
class: region.body_region | status: RESOLVED | executor: fma_mesh
identity: {"fma_id":"FMA7154","kind":"fma_concept","part_ids":["FJ1252","FJ1253","FJ1282","FJ1285","FJ1286","FJ1289","FJ1294","FJ1295","FJ1297","FJ1298","FJ1299","FJ1301","FJ1302","FJ1304","FJ1305","FJ1306","FJ1308","FJ1309","FJ1317","FJ1320","FJ1322","FJ1323","FJ1324","FJ1328","FJ1331","FJ1336","FJ1337","FJ1340","FJ1345","FJ1346","FJ1348","FJ1349","FJ1350","FJ1352","FJ1353","FJ1355","FJ1356","FJ1357","FJ1359","FJ1360","FJ1368","FJ1371","FJ1373","FJ1374","FJ1375","FJ1379","FJ1382","FJ1730","FJ1731","FJ1732","FJ1733","FJ1738","FJ1739","FJ1740","FJ1743","FJ1744","FJ1745","FJ1746","FJ1747","FJ1748","FJ1749","FJ1750","FJ1751","FJ1758","FJ1759","FJ1760","FJ1762","FJ1767","FJ1769","FJ1770","FJ1775","FJ1779","FJ1780","FJ1781","FJ1783","FJ1784","FJ1785","FJ1786","FJ1787","FJ1788","FJ1789","FJ1790","FJ1791","FJ1792","FJ1795","FJ1797","FJ1798","FJ1800","FJ1801","FJ1806","FJ1807","FJ1808","FJ1810","FJ1814","FJ1817","FJ1822","FJ1826","FJ1828","FJ1830","FJ1831","FJ1833","FJ1834","FJ1835","FJ1836","FJ1841","FJ1842","FJ2554","FJ2555","FJ2556","FJ2557","FJ2558","FJ2741","FJ2748","FJ2753","FJ2760","FJ2761","FJ2762","FJ2766","FJ2768","FJ2772","FJ3199","FJ3200","FJ3201","FJ3263","FJ3265","FJ3269","FJ3272","FJ3273","FJ3274","FJ3281","FJ3287","FJ3289","FJ3309","FJ3369","FJ3371","FJ3375","FJ3378","FJ3379","FJ3380","FJ3386","FJ3392","FJ3394","FJ3395"]}
reason: None

LM:S:GB14:location:72-78
source: centre
class: None | status: UNRESOLVED | executor: explicit_unresolved_taxonomy
identity: {"kind":null}
reason: source_backed_derived_unresolved

LM:S:GB14:location:82-91
source: the pupil
class: soft_tissue_feature | status: UNRESOLVED | executor: atlas_name_resolution
identity: {"kind":null}
reason: no unique atlas name match

Relation / operator chain

RL:S:GB14:location:00:relative-to
status: RESOLVED | operator: directional_relation
operand IDs: LM:S:GB14:location:32-45
direction: superior | source statement: S:GB14:location
reason: None

RL:S:GB14:location:01:relative-to
status: UNRESOLVED | operator: directional_relation
operand IDs: LM:S:GB14:location:67-91
direction: None | source statement: None
reason: argument landmark not uniquely executable

RL:S:GB14:location:02:center-of
status: UNRESOLVED | operator: entity_center
operand IDs: LM:S:GB14:location:82-91
direction: None | source statement: None
reason: argument landmark not uniquely executable

RL:S:GB14:location:03:surface-landmark
status: RESOLVED | operator: surface_membership
operand IDs: LM:S:GB14:location:7-11
direction: None | source statement: S:GB14:location
reason: None

Measurements / calibration use

{"anchor_landmark_id":"LM:S:GB14:location:32-45","direction":"superior","executor":"approved_calibration","kind":"proportional_offset","measurement_id":"PM:S:GB14:location:00","reason":null,"source_statement_id":"S:GB14:location","source_unit":"B-cun","status":"RESOLVED","unit":"B-cun","value":1.0,"who_pdf_page":187}

Dependency edges (projection of existing explicit IDs)

LM:S:GB14:location:32-45 -> PM:S:GB14:location:00 [anchor_to_measurement]

PM:S:GB14:location:00 -> S:GB14:location [measurement_to_statement]

LM:S:GB14:location:32-45 -> RL:S:GB14:location:00:relative-to [operand_to_relation]
LM:S:GB14:location:67-91 -> RL:S:GB14:location:01:relative-to [operand_to_relation]
LM:S:GB14:location:7-11 -> RL:S:GB14:location:03:surface-landmark [operand_to_relation]
LM:S:GB14:location:82-91 -> RL:S:GB14:location:02:center-of [operand_to_relation]
RL:S:GB14:location:00:relative-to -> S:GB14:location [relation_to_statement]
RL:S:GB14:location:03:surface-landmark -> S:GB14:location [relation_to_statement]
S:GB14:location -> GB14 [statement_to_point]

Recorded data order

Stage 2 output does not record timestamped per-operation runtime order. The following preserves exact serialized Stage 2 data order.

LM:S:GB14:location:32-45 -> LM:S:GB14:location:67-91 -> LM:S:GB14:location:7-11 -> LM:S:GB14:location:72-78 -> LM:S:GB14:location:82-91 ->
PM:S:GB14:location:00 -> RL:S:GB14:location:00:relative-to -> RL:S:GB14:location:01:relative-to -> RL:S:GB14:location:02:center-of ->
RL:S:GB14:location:03:surface-landmark -> S:GB14:location -> GB14

Final semantic execution result

{ "location_statement_status": "UNRESOLVED", "primary_location_status": "UNRESOLVED" }

Blocking dependency

{ "blocking_items": [{ "landmark_class": "soft_tissue_feature", "node_id": "LM:S:GB14:location:67-91", "node_type": "landmark", "reason": "no unique atlas name match", "source_raw": " the centre of the pupil", "status": "UNRESOLVED" }, { "landmark_class": "soft_tissue_feature", "node_id": "LM:S:GB14:location:82-91", "node_type": "landmark", "reason": "no unique atlas name match", "source_raw": "the pupil", "status": "UNRESOLVED" }, { "downstream_stop_items": ["RL:S:GB14:location:01:relative-to", "RL:S:GB14:location:02:center-of"], "first_unresolved_dependency": { "landmark_class": "soft_tissue_feature", "node_id": "LM:S:GB14:location:67-91", "node_type": "landmark", "reason": "no unique atlas name match", "source_raw": " the centre of the pupil", "status": "UNRESOLVED" }, "upstream_resolved_dependencies": ["RL:S:GB14:location:00:relative-to", "LM:S:GB14:location:32-45", "RL:S:GB14:location:03:surface-landmark", "LM:S:GB14:location:7-11"] }] }

Coordinates generated: 0 | Legacy C coordinate referenced: 0

- [] semantic decomposition correct
- [] operator semantics correct
- [] operand identity correct
- [] dependency structure correct
- [] unresolved/conditional handling correct
- [] ACCEPT
- [] REJECT

LU6 - UNRESOLVED

WHO primary Location

S:LU6:location: On the anterolateral aspect of the forearm, on the line connecting LU5 with LU9, 7 B-cun superior to the palmar wrist crease.

Semantic decomposition / operands

LM:S:LU6:location:100-124
source: the palmar wrist crease
class: crease.skin_crease | status: RESOLVED | executor: approved_surface_registry
identity: {"geometry_hash":"383e200f286d70d3c13689830673e15bb470bf299882294478ffe17ce465fa2f","kind":"surface_registry","registry_id":"SR:palmar_wrist_crease"}
reason: None

LM:S:LU6:location:51-66
source: line connecting
class: line.reference_line | status: UNRESOLVED | executor: no_family_executor
identity: {"kind":null}
reason: unsupported landmark disposition/class: geometry_or_surface/line.reference_line

LM:S:LU6:location:67-70
source: LU5
class: acupoint_reference | status: RESOLVED | executor: reference_acupoint
identity: {"kind":"reference_acupoint","reference_point_id":"LU5"}
reason: None

LM:S:LU6:location:7-27
source: anterolateral aspect
class: surface.aspect | status: RESOLVED | executor: approved_surface_registry
identity: {"geometry_hash":"8ce43d41eaba7ffc905ab399e621546e96abede4af22e20655f5814a9f8b665c","kind":"surface_registry","registry_id":"SR:anterolateral_forearm"}
reason: None

LM:S:LU6:location:76-79
source: LU9
class: acupoint_reference | status: RESOLVED | executor: reference_acupoint
identity: {"kind":"reference_acupoint","reference_point_id":"LU9"}
reason: None

LM:S:LU6:note:1:103-106
source: LU9
class: acupoint_reference | status: RESOLVED | executor: reference_acupoint
identity: {"kind":"reference_acupoint","reference_point_id":"LU9"}
reason: None

LM:S:LU6:note:1:32-36
source: LU5
class: acupoint_reference | status: RESOLVED | executor: reference_acupoint
identity: {"kind":"reference_acupoint","reference_point_id":"LU5"}
reason: None

LM:S:LU6:note:1:57-106
source: the midpoint of the line connecting LU5 with LU9
class: line.reference_line | status: UNRESOLVED | executor: no_family_executor
identity: {"kind":null}
reason: unsupported landmark disposition/class: geometry_or_surface/line.reference_line

LM:S:LU6:note:1:94-97
source: LU5
class: acupoint_reference | status: RESOLVED | executor: reference_acupoint
identity: {"kind":"reference_acupoint","reference_point_id":"LU5"}
reason: None

LM21:S:LU6:location:67-70:002c0a95e5
source: LU5
class: acupoint_reference | status: RESOLVED | executor: reference_acupoint
identity: {"kind":"reference_acupoint","reference_point_id":"LU5"}
reason: None

LM21:S:LU6:location:76-79:30c4b43ed7
source: LU9
class: acupoint_reference | status: RESOLVED | executor: reference_acupoint
identity: {"kind":"reference_acupoint","reference_point_id":"LU9"}
reason: None

Relation / operator chain

RL:S:LU6:location:00:on-line
status: RESOLVED | operator: line_membership
operand IDs: LM:S:LU6:location:67-70, LM:S:LU6:location:76-79
direction: None | source statement: S:LU6:location
reason: None

RL:S:LU6:location:01:relative-to
status: RESOLVED | operator: directional_relation
operand IDs: LM:S:LU6:location:100-124
direction: superior | source statement: S:LU6:location
reason: None

RL:S:LU6:location:02:surface-landmark
status: RESOLVED | operator: surface_membership
operand IDs: LM:S:LU6:location:7-27
direction: None | source statement: S:LU6:location
reason: None

RL:S:LU6:location:03:surface-landmark
status: UNRESOLVED | operator: surface_membership
operand IDs: LM:S:LU6:location:51-66
direction: None | source statement: None
reason: argument landmark not uniquely executable

RL:S:LU6:note:1:00:on-line
status: RESOLVED | operator: line_membership
operand IDs: LM:S:LU6:note:1:94-97, LM:S:LU6:note:1:103-106
direction: None | source statement: S:LU6:note:1
reason: None

RL:S:LU6:note:1:01:midpoint-between
status: RESOLVED | operator: midpoint_between
operand IDs: LM:S:LU6:note:1:94-97, LM:S:LU6:note:1:103-106
direction: None | source statement: S:LU6:note:1
reason: None

RL:S:LU6:note:1:02:relative-to
status: RESOLVED | operator: directional_relation
operand IDs: LM:S:LU6:note:1:32-36
direction: inferior | source statement: S:LU6:note:1
reason: None

RL:S:LU6:note:1:03:relative-to
status: UNRESOLVED | operator: directional_relation
operand IDs: LM:S:LU6:note:1:57-106
direction: None | source statement: None
reason: argument landmark not uniquely executable

RL21:S:LU6:location:67-70:reference-acupoint:f42e2db8d9
status: RESOLVED | operator: reference_dependency
operand IDs: LM21:S:LU6:location:67-70:002c0a95e5
direction: None | source statement: S:LU6:location
reason: None

RL21:S:LU6:location:76-79:reference-acupoint:5171900d45
status: RESOLVED | operator: reference_dependency
operand IDs: LM21:S:LU6:location:76-79:30c4b43ed7
direction: None | source statement: S:LU6:location
reason: None

RL21:S:LU6:note:1:33-36:reference-acupoint:a79a726278
status: RESOLVED | operator: reference_dependency
operand IDs: LM:S:LU6:note:1:32-36
direction: None | source statement: S:LU6:note:1
reason: None

RL21:S:LU6:note:1:94-97:reference-acupoint:0d3fedaf0b
status: RESOLVED | operator: reference_dependency
operand IDs: LM:S:LU6:note:1:94-97
direction: None | source statement: S:LU6:note:1
reason: None

RL21:S:LU6:note:1:103-106:reference-acupoint:ffaf5ca270
status: RESOLVED | operator: reference_dependency
operand IDs: LM:S:LU6:note:1:103-106
direction: None | source statement: S:LU6:note:1
reason: None

Measurements / calibration use

```
{"anchor_landmark_id":"LM:S:LU6:location:100-124","direction":"superior","executor":"approved_calibration","kind":"proportional_offset","measurement_id":"PM:S:LU6:location:00","reason":null,"source_statement_id":"S:LU6:location","source_unit":"B-cun","status":"RESOLVED","unit":"B-cun","value":7.0,"who_pdf_page":37}
```

```
{"anchor_landmark_id":"LM:S:LU6:note:1:32-36","direction":"inferior","executor":"approved_calibration","kind":"proportional_offset","measurement_id":"PM:S:LU6:note:1:00","reason":null,"source_statement_id":"S:LU6:note:1","source_unit":"B-cun","status":"RESOLVED","unit":"B-cun","value":5.0,"who_pdf_page":37}

{"anchor_landmark_id":"LM:S:LU6:note:1:57-106","direction":"superior","executor":"approved_calibration","kind":"proportional_offset","measurement_id":"PM:S:LU6:note:1:01","reason":null,"source_statement_id":"S:LU6:note:1","source_unit":"B-cun","status":"RESOLVED","unit":"B-cun","value":1.0,"who_pdf_page":37}
```

Dependency edges (projection of existing explicit IDs)

```
LM:S:LU6:location:100-124 -> PM:S:LU6:location:00 [anchor_to_measurement]
LM:S:LU6:note:1:32-36 -> PM:S:LU6:note:1:00 [anchor_to_measurement]
LM:S:LU6:note:1:57-106 -> PM:S:LU6:note:1:01 [anchor_to_measurement]
PM:S:LU6:location:00 -> S:LU6:location [measurement_to_statement]
PM:S:LU6:note:1:00 -> S:LU6:note:1 [measurement_to_statement]
PM:S:LU6:note:1:01 -> S:LU6:note:1 [measurement_to_statement]
LM21:S:LU6:location:67-70:002c0a95e5 -> RL21:S:LU6:location:67-70:reference-acupoint:f42e2db8d9 [operand_to_relation]
LM21:S:LU6:location:76-79:30c4b43ed7 -> RL21:S:LU6:location:76-79:reference-acupoint:5171900d45 [operand_to_relation]
LM:S:LU6:location:100-124 -> RL:S:LU6:location:01:relative-to [operand_to_relation]
LM:S:LU6:location:51-66 -> RL:S:LU6:location:03:surface-landmark [operand_to_relation]
LM:S:LU6:location:67-70 -> RL:S:LU6:location:00:on-line [operand_to_relation]
LM:S:LU6:location:7-27 -> RL:S:LU6:location:02:surface-landmark [operand_to_relation]
LM:S:LU6:location:76-79 -> RL:S:LU6:location:00:on-line [operand_to_relation]
LM:S:LU6:note:1:103-106 -> RL21:S:LU6:note:1:103-106:reference-acupoint:ffaf5ca270 [operand_to_relation]
LM:S:LU6:note:1:103-106 -> RL:S:LU6:note:1:00:on-line [operand_to_relation]
LM:S:LU6:note:1:103-106 -> RL:S:LU6:note:1:01:midpoint-between [operand_to_relation]
LM:S:LU6:note:1:32-36 -> RL21:S:LU6:note:1:33-36:reference-acupoint:a79a726278 [operand_to_relation]
LM:S:LU6:note:1:32-36 -> RL:S:LU6:note:1:02:relative-to [operand_to_relation]
LM:S:LU6:note:1:57-106 -> RL:S:LU6:note:1:03:relative-to [operand_to_relation]
LM:S:LU6:note:1:94-97 -> RL21:S:LU6:note:1:94-97:reference-acupoint:0d3fedaf0b [operand_to_relation]
LM:S:LU6:note:1:94-97 -> RL:S:LU6:note:1:00:on-line [operand_to_relation]
LM:S:LU6:note:1:94-97 -> RL:S:LU6:note:1:01:midpoint-between [operand_to_relation]
RL21:S:LU6:location:67-70:reference-acupoint:f42e2db8d9 -> S:LU6:location [relation_to_statement]
RL21:S:LU6:location:76-79:reference-acupoint:5171900d45 -> S:LU6:location [relation_to_statement]
RL21:S:LU6:note:1:103-106:reference-acupoint:ffaf5ca270 -> S:LU6:note:1 [relation_to_statement]
RL21:S:LU6:note:1:33-36:reference-acupoint:a79a726278 -> S:LU6:note:1 [relation_to_statement]
RL21:S:LU6:note:1:94-97:reference-acupoint:0d3fedaf0b -> S:LU6:note:1 [relation_to_statement]
RL:S:LU6:location:00:on-line -> S:LU6:location [relation_to_statement]
RL:S:LU6:location:01:relative-to -> S:LU6:location [relation_to_statement]
RL:S:LU6:location:02:surface-landmark -> S:LU6:location [relation_to_statement]
RL:S:LU6:note:1:00:on-line -> S:LU6:note:1 [relation_to_statement]
RL:S:LU6:note:1:01:midpoint-between -> S:LU6:note:1 [relation_to_statement]
RL:S:LU6:note:1:02:relative-to -> S:LU6:note:1 [relation_to_statement]
S:LU6:location -> LU6 [statement_to_point]
S:LU6:note:1 -> LU6 [statement_to_point]
```

Recorded data order

Stage 2 output does not record timestamped per-operation runtime order. The following preserves exact serialized Stage 2 data order.

```
LM:S:LU6:location:100-124 -> LM:S:LU6:location:51-66 -> LM:S:LU6:location:67-70 -> LM:S:LU6:location:7-27 -> LM:S:LU6:location:76-79 ->
LM:S:LU6:note:1:103-106 -> LM:S:LU6:note:1:32-36 -> LM:S:LU6:note:1:57-106 -> LM:S:LU6:note:1:94-97 -> LM21:S:LU6:location:67-70:002c0a95e5
-> LM21:S:LU6:location:76-79:30c4b43ed7 -> PM:S:LU6:location:00 -> PM:S:LU6:note:1:00 -> PM:S:LU6:note:1:01 -> RL:S:LU6:location:00:on-line ->
RL:S:LU6:location:01:relative-to -> RL:S:LU6:location:02:surface-landmark -> RL:S:LU6:location:03:surface-landmark -> RL:S:LU6:note:1:00:on-line ->
RL:S:LU6:note:1:01:midpoint-between -> RL:S:LU6:note:1:02:relative-to -> RL:S:LU6:note:1:03:relative-to ->
RL21:S:LU6:location:67-70:reference-acupoint:f42e2db8d9 -> RL21:S:LU6:location:76-79:reference-acupoint:5171900d45 ->
RL21:S:LU6:note:1:33-36:reference-acupoint:a79a726278 -> RL21:S:LU6:note:1:94-97:reference-acupoint:0d3fedaf0b ->
RL21:S:LU6:note:1:103-106:reference-acupoint:ffaf5ca270 -> S:LU6:location -> S:LU6:note:1 -> LU6
```

Final semantic execution result

```
{"location_statement_status":"UNRESOLVED","primary_location_status":"UNRESOLVED"}
```

Blocking dependency

```
{ "blocking_items": [ { "landmark_class": "line.reference_line", "node_id": "LM:S:LU6:location:51-66", "node_type": "landmark", "reason": "unsupported landmark disposition/class: geometry_or_surface/line.reference_line", "source_raw": "line connecting", "status": "UNRESOLVED" } ], "downstream_stop_items": [ "RL:S:LU6:location:03:surface-landmark" ], "first_unresolved_dependency": { "landmark_class": "line.reference_line", "node_id": "LM:S:LU6:location:51-66", "node_type": "landmark", "reason": "unsupported landmark disposition/class: geometry_or_surface/line.reference_line", "source_raw": "line connecting", "status": "UNRESOLVED" }, "upstream_resolved_dependencies": [ "RL:S:LU6:location:00:on-line", "LM:S:LU6:location:67-70", "LM:S:LU6:location:76-79", "RL:S:LU6:location:01:relative-to", "LM:S:LU6:location:100-124", "RL:S:LU6:location:02:surface-landmark", "LM:S:LU6:location:7-27", "RL21:S:LU6:location:67-70:reference-acupoint:f42e2db8d9", "LM21:S:LU6:location:67-70:002c0a95e5", "RL21:S:LU6:location:76-79:reference-acupoint:5171900d45", "LM21:S:LU6:location:76-79:30c4b43ed7" ] }
```

Coordinates generated: 0 | Legacy C coordinate referenced: 0

- [] semantic decomposition correct
- [] operator semantics correct
- [] operand identity correct
- [] dependency structure correct
- [] unresolved/conditional handling correct
- [] ACCEPT
- [] REJECT

LI7 - UNRESOLVED

WHO primary Location

S:LI7:location: On the posterolateral aspect of the forearm, on the line connecting LI5 with LI11, 5 B-cun superior to the dorsal wrist crease.

Semantic decomposition / operands

LM:S:LI7:location:102-126
source: the dorsal wrist crease
class: crease.skin_crease | status: RESOLVED | executor: approved_surface_registry
identity: {"geometry_hash":"0ff534739841ae12b463e9ea5102d7801e5de256871212f7ddeb84a0b0265288","kind":"surface_registry","registry_id":"SR:dorsal_wrist_crease"}
reason: None

LM:S:LI7:location:52-67
source: line connecting
class: line.reference_line | status: UNRESOLVED | executor: no_family_executor
identity: {"kind":null}
reason: unsupported landmark disposition/class: geometry_or_surface/line.reference_line

LM:S:LI7:location:68-71
source: LI5
class: acupoint_reference | status: RESOLVED | executor: reference_acupoint
identity: {"kind":"reference_acupoint","reference_point_id":"LI5"}
reason: None

LM:S:LI7:location:7-28
source: posterolateral aspect
class: surface.aspect | status: RESOLVED | executor: approved_surface_registry
identity: {"geometry_hash":"4e542fb31cd71e7a148b3607b7424ea18ab74b0b8ddb0210ce3c5abccc2c54d9","kind":"surface_registry","registry_id":"SR:posterolateral_forearm"}
reason: None

LM:S:LI7:location:77-81
source: LI11
class: acupoint_reference | status: RESOLVED | executor: reference_acupoint
identity: {"kind":"reference_acupoint","reference_point_id":"LI11"}
reason: None

LM21:S:LI7:location:68-71:1bff496df7
source: LI5
class: acupoint_reference | status: RESOLVED | executor: reference_acupoint
identity: {"kind":"reference_acupoint","reference_point_id":"LI5"}
reason: None

LM21:S:LI7:location:77-81:245ea01482
source: LI11
class: acupoint_reference | status: RESOLVED | executor: reference_acupoint
identity: {"kind":"reference_acupoint","reference_point_id":"LI11"}
reason: None

Relation / operator chain

RL:S:LI7:location:00:on-line
status: RESOLVED | operator: line_membership
operand IDs: LM:S:LI7:location:68-71, LM:S:LI7:location:77-81
direction: None | source statement: S:LI7:location
reason: None

RL:S:LI7:location:01:relative-to
status: RESOLVED | operator: directional_relation
operand IDs: LM:S:LI7:location:102-126
direction: superior | source statement: S:LI7:location
reason: None

RL:S:LI7:location:02:surface-landmark
status: RESOLVED | operator: surface_membership
operand IDs: LM:S:LI7:location:7-28
direction: None | source statement: S:LI7:location
reason: None

RL:S:LI7:location:03:surface-landmark
status: UNRESOLVED | operator: surface_membership
operand IDs: LM:S:LI7:location:52-67
direction: None | source statement: None
reason: argument landmark not uniquely executable

RL21:S:LI7:location:68-71:reference-acupoint:931d5b1cfd
status: RESOLVED | operator: reference_dependency
operand IDs: LM21:S:LI7:location:68-71:1bff496df7
direction: None | source statement: S:LI7:location
reason: None

RL21:S:LI7:location:77-81:reference-acupoint:bbace3ac1c
status: RESOLVED | operator: reference_dependency
operand IDs: LM21:S:LI7:location:77-81:245ea01482
direction: None | source statement: S:LI7:location
reason: None

Measurements / calibration use

```
{"anchor_landmark_id":"LM:S:LI7:location:102-126","direction":"superior","executor":"approved_calibration","kind":"proportional_offset","measurement_id":"PM:S:LI7:location:00","reason":null,"source_statement_id":"S:LI7:location","source_unit":"B-cun","status":"RESOLVED","unit":"B-cun","value":5.0,"who_pdf_page":46}
```

Dependency edges (projection of existing explicit IDs)

LM:S:LI7:location:102-126 -> PM:S:LI7:location:00 [anchor_to_measurement]
PM:S:LI7:location:00 -> S:LI7:location [measurement_to_statement]
LM21:S:LI7:location:68-71:1bff496df7 -> RL21:S:LI7:location:68-71:reference-acupoint:931d5b1cfd [operand_to_relation]
LM21:S:LI7:location:77-81:245ea01482 -> RL21:S:LI7:location:77-81:reference-acupoint:bbace3ac1c [operand_to_relation]
LM:S:LI7:location:102-126 -> RL:S:LI7:location:01:relative-to [operand_to_relation]
LM:S:LI7:location:52-67 -> RL:S:LI7:location:03:surface-landmark [operand_to_relation]
LM:S:LI7:location:68-71 -> RL:S:LI7:location:00:on-line [operand_to_relation]
LM:S:LI7:location:7-28 -> RL:S:LI7:location:02:surface-landmark [operand_to_relation]
LM:S:LI7:location:77-81 -> RL:S:LI7:location:00:on-line [operand_to_relation]
RL21:S:LI7:location:68-71:reference-acupoint:931d5b1cfd -> S:LI7:location [relation_to_statement]
RL21:S:LI7:location:77-81:reference-acupoint:bbace3ac1c -> S:LI7:location [relation_to_statement]
RL:S:LI7:location:00:on-line -> S:LI7:location [relation_to_statement]
RL:S:LI7:location:01:relative-to -> S:LI7:location [relation_to_statement]
RL:S:LI7:location:02:surface-landmark -> S:LI7:location [relation_to_statement]
S:LI7:location -> LI7 [statement_to_point]

Recorded data order

Stage 2 output does not record timestamped per-operation runtime order. The following preserves exact serialized Stage 2 data order.

LM:S:LI7:location:102-126 -> LM:S:LI7:location:52-67 -> LM:S:LI7:location:68-71 -> LM:S:LI7:location:7-28 -> LM:S:LI7:location:77-81 -> LM21:S:LI7:location:68-71:1bff496df7 -> LM21:S:LI7:location:77-81:245ea01482 -> PM:S:LI7:location:00 -> RL:S:LI7:location:00:on-line -> RL:S:LI7:location:01:relative-to -> RL:S:LI7:location:02:surface-landmark -> RL:S:LI7:location:03:surface-landmark -> RL21:S:LI7:location:68-71:reference-acupoint:931d5b1cfd -> RL21:S:LI7:location:77-81:reference-acupoint:bbace3ac1c -> S:LI7:location -> LI7

Final semantic execution result

```
{"location_statement_status":"UNRESOLVED","primary_location_status":"UNRESOLVED"}
```

Blocking dependency

```
{"blocking_items":[{"landmark_class":"line.reference_line","node_id":"LM:S:LI7:location:52-67","node_type":"landmark","reason":"unsupported landmark disposition/class: geometry_or_surface/line.reference_line","source_raw":"line connecting","status":"UNRESOLVED"}],"downstream_stop_items":["RL:S:LI7:location:03:surface-landmark"],"first_unresolved_dependency":{"landmark_class":"line.reference_line","node_id":"LM:S:LI7:location:52-67","node_type":"landmark","reason":"unsupported landmark disposition/class: geometry_or_surface/line.reference_line","source_raw":"line connecting","status":"UNRESOLVED"},"upstream_resolved_dependencies":{"RL:S:LI7:location:00:on-line","LM:S:LI7:location:68-71","LM:S:LI7:location:77-81","RL:S:LI7:location:01:relative-to","LM:S:LI7:location:102-126","RL:S:LI7:location:02:surface-landmark","LM:S:LI7:location:7-28","RL21:S:LI7:location:68-71:reference-acupoint:931d5b1cfd","LM21:S:LI7:location:68-71:1bff496df7","RL21:S:LI7:location:77-81:reference-acupoint:bbace3ac1c","LM21:S:LI7:location:77-81:245ea01482"}}
```

Coordinates generated: 0 | Legacy C coordinate referenced: 0

[] semantic decomposition correct
[] operator semantics correct
[] operand identity correct
[] dependency structure correct
[] unresolved/conditional handling correct
[] ACCEPT
[] REJECT

GB26 - UNRESOLVED

WHO primary Location

S:GB26:location: On the lateral abdomen, inferior to the free extremity of the 11th rib, at the same level as the centre of umbilicus.

Semantic decomposition / operands

LM:S:GB26:location:107-116

source: umbilicus

class: orifice_or_cavity | status: UNRESOLVED | executor: atlas_name_resolution

identity: {"kind":null}

reason: no unique atlas name match

LM:S:GB26:location:35-70

source: the free extremity of the 11th rib

class: structure.part | status: UNRESOLVED | executor: atlas_name_resolution

identity: {"kind":null}

reason: no unique atlas name match

LM:S:GB26:location:40-54

source: free extremity

class: structure.part | status: RESOLVED | executor: composite_subfeature

identity: {"kind":"entity_subfeature","modifier":"free","parent_landmark_id":"LM:S:GB26:location:58-70"}

reason: None

LM:S:GB26:location:58-70

source: the 11th rib

class: bone.bone | status: RESOLVED | executor: fma_mesh

identity: {"fma_id":"FMA8499","kind":"fma_concept","part_ids":["FJ3226","FJ3331"]}

reason: None

LM:S:GB26:location:7-22

source: lateral abdomen

class: region.body_region | status: RESOLVED | executor: approved_surface_registry

identity: {"geometry_hash":"912678a19ed1d94961027afd11c1d686ebc1dd7d67fe77a92a26935f99509145","kind":"surface_registry","registry_id":"SR:lateral_abdomen"}

reason: None

LM:S:GB26:location:93-116

source: the centre of umbilicus

class: orifice_or_cavity | status: UNRESOLVED | executor: atlas_name_resolution

identity: {"kind":null}

reason: no unique atlas name match

LM:S:GB26:location:97-103

source: centre

class: None | status: UNRESOLVED | executor: explicit_unresolved_taxonomy

identity: {"kind":null}

reason: source_backed_derived_unresolved

LM:S:GB26:note:1:103-189

source: the 11th rib that is situated immediately below the inferior border of the costal arch

class: boundary.border | status: UNRESOLVED | executor: no_family_executor

identity: {"kind":null}

reason: unsupported landmark disposition/class: geometry_or_surface/boundary.border

LM:S:GB26:note:1:85-99

source: free extremity

class: structure.part | status: UNRESOLVED | executor: atlas_name_resolution

identity: {"kind":null}

reason: no unique atlas name match

LM:S:GB26:note:2:27-32

source: LR13

class: acupoint_reference | status: RESOLVED | executor: reference_acupoint

identity: {"kind":"reference_acupoint","reference_point_id":"LR13"}

reason: None

LM:S:GB26:note:2:55-58

source: CV8

class: acupoint_reference | status: RESOLVED | executor: reference_acupoint

identity: {"kind":"reference_acupoint","reference_point_id":"CV8"}

reason: None

Relation / operator chain

RL:S:GB26:location:00:relative-to

status: UNRESOLVED | operator: directional_relation

operand IDs: LM:S:GB26:location:35-70

direction: None | source statement: None

reason: argument landmark not uniquely executable

RL:S:GB26:location:01:same-level
status: UNRESOLVED | operator: same_level_plane
operand IDs: LM:S:GB26:location:93-116
direction: None | source statement: None
reason: argument landmark not uniquely executable

RL:S:GB26:location:02:center-of
status: UNRESOLVED | operator: entity_center
operand IDs: LM:S:GB26:location:107-116
direction: None | source statement: None
reason: argument landmark not uniquely executable

RL:S:GB26:location:03:surface-landmark
status: RESOLVED | operator: surface_membership
operand IDs: LM:S:GB26:location:7-22
direction: None | source statement: S:GB26:location
reason: None

RL:S:GB26:note:2:00:relative-to
status: RESOLVED | operator: directional_relation
operand IDs: LM:S:GB26:note:2:27-32
direction: inferior | source statement: S:GB26:note:2
reason: None

RL:S:GB26:note:2:01:same-level
status: RESOLVED | operator: same_level_plane
operand IDs: LM:S:GB26:note:2:55-58
direction: None | source statement: S:GB26:note:2
reason: None

RL21:S:GB26:note:2:28-32:reference-acupoint:260f7e9589
status: RESOLVED | operator: reference_dependency
operand IDs: LM:S:GB26:note:2:27-32
direction: None | source statement: S:GB26:note:2
reason: None

RL21:S:GB26:note:2:55-58:reference-acupoint:3e3fd2558d
status: RESOLVED | operator: reference_dependency
operand IDs: LM:S:GB26:note:2:55-58
direction: None | source statement: S:GB26:note:2
reason: None

Dependency edges (projection of existing explicit IDs)

LM:S:GB26:location:107-116 -> RL:S:GB26:location:02:center-of [operand_to_relation]
LM:S:GB26:location:35-70 -> RL:S:GB26:location:00:relative-to [operand_to_relation]
LM:S:GB26:location:7-22 -> RL:S:GB26:location:03:surface-landmark [operand_to_relation]
LM:S:GB26:location:93-116 -> RL:S:GB26:location:01:same-level [operand_to_relation]
LM:S:GB26:note:2:27-32 -> RL21:S:GB26:note:2:28-32:reference-acupoint:260f7e9589 [operand_to_relation]
LM:S:GB26:note:2:27-32 -> RL:S:GB26:note:2:00:relative-to [operand_to_relation]
LM:S:GB26:note:2:55-58 -> RL21:S:GB26:note:2:55-58:reference-acupoint:3e3fd2558d [operand_to_relation]
LM:S:GB26:note:2:55-58 -> RL:S:GB26:note:2:01:same-level [operand_to_relation]
RL21:S:GB26:note:2:28-32:reference-acupoint:260f7e9589 -> S:GB26:note:2 [relation_to_statement]
RL21:S:GB26:note:2:55-58:reference-acupoint:3e3fd2558d -> S:GB26:note:2 [relation_to_statement]
RL:S:GB26:location:03:surface-landmark -> S:GB26:location [relation_to_statement]
RL:S:GB26:note:2:00:relative-to -> S:GB26:note:2 [relation_to_statement]
RL:S:GB26:note:2:01:same-level -> S:GB26:note:2 [relation_to_statement]
S:GB26:location -> GB26 [statement_to_point]
S:GB26:note:1 -> GB26 [statement_to_point]
S:GB26:note:2 -> GB26 [statement_to_point]

Recorded data order

Stage 2 output does not record timestamped per-operation runtime order. The following preserves exact serialized Stage 2 data order.

LM:S:GB26:location:107-116 -> LM:S:GB26:location:35-70 -> LM:S:GB26:location:40-54 -> LM:S:GB26:location:58-70 -> LM:S:GB26:location:7-22 -> LM:S:GB26:location:93-116 -> LM:S:GB26:location:97-103 -> LM:S:GB26:note:1:103-189 -> LM:S:GB26:note:1:85-99 -> LM:S:GB26:note:2:27-32 -> LM:S:GB26:note:2:55-58 -> RL:S:GB26:location:00:relative-to -> RL:S:GB26:location:01:same-level -> RL:S:GB26:location:02:center-of -> RL:S:GB26:location:03:surface-landmark -> RL:S:GB26:note:2:00:relative-to -> RL:S:GB26:note:2:01:same-level -> RL21:S:GB26:note:2:28-32:reference-acupoint:260f7e9589 -> RL21:S:GB26:note:2:55-58:reference-acupoint:3e3fd2558d -> S:GB26:location -> S:GB26:note:1 -> S:GB26:note:2 -> GB26

Final semantic execution result

```
{"location_statement_status":"UNRESOLVED","primary_location_status":"UNRESOLVED"}
```

Blocking dependency

```
{
  "blocking_items": [
    {
      "landmark_class": "structure.part",
      "node_id": "LM:S:GB26:location:35-70",
      "node_type": "landmark",
      "reason": "no unique atlas name match",
      "source_raw": " the free extremity of the 11th rib",
      "status": "UNRESOLVED"
    },
    {
      "landmark_class": "orifice_or_cavity",
      "node_id": "LM:S:GB26:location:93-116",
      "node_type": "landmark",
      "reason": "no unique atlas name match",
      "source_raw": "the centre of umbilicus",
      "status": "UNRESOLVED"
    },
    {
      "landmark_class": "orifice_or_cavity",
      "node_id": "LM:S:GB26:location:107-116",
      "node_type": "landmark",
      "reason": "no unique atlas name match",
      "source_raw": "umbilicus",
      "status": "UNRESOLVED"
    }
  ],
  "downstream_stop_items": [
    {
      "RL": "S:GB26:location:00:relative-to",
      "RL": "S:GB26:location:01:same-level",
      "RL": "S:GB26:location:02:center-of",
      "first_unresolved_dependency": {
        "landmark_class": "structure.part",
        "node_id": "LM:S:GB26:location:35-70",
        "node_type": "landmark",
        "reason": "no unique atlas name match",
        "source_raw": " the free extremity of the 11th rib",
        "status": "UNRESOLVED"
      },
      "upstream_resolved_dependencies": [
        {
          "RL": "S:GB26:location:03:surface-landmark",
          "LM": "S:GB26:location:7-22"
        }
      ]
    }
  ]
}
```

Coordinates generated: 0 | Legacy C coordinate referenced: 0

- [] semantic decomposition correct
- [] operator semantics correct
- [] operand identity correct
- [] dependency structure correct
- [] unresolved/conditional handling correct
- [] ACCEPT
- [] REJECT

ST2 - UNRESOLVED

WHO primary Location

S:ST2:location: On the face, in the infraorbital foramen.

Semantic decomposition / operands

LM:S:ST2:location:20-40
source: infraorbital foramen
class: bone.opening | status: UNRESOLVED | executor: atlas_name_resolution
identity: {"kind":null}
reason: no unique atlas name match

LM:S:ST2:location:7-11
source: face
class: region.body_region | status: RESOLVED | executor: fma_mesh
identity: {"fma_id":"FMA24728","kind":"fma_concept","part_ids":["FJ1252","FJ1253","FJ1282","FJ1285","FJ1286","FJ1289","FJ1294","FJ1295","FJ1297","FJ1298","FJ1299","FJ1301","FJ1302","FJ1304","FJ1305","FJ1306","FJ1308","FJ1309","FJ1317","FJ1320","FJ1322","FJ1323","FJ1324","FJ1328","FJ1331","FJ1336","FJ1337","FJ1340","FJ1345","FJ1346","FJ1348","FJ1349","FJ1350","FJ1352","FJ1353","FJ1355","FJ1356","FJ1357","FJ1359","FJ1360","FJ1368","FJ1371","FJ1373","FJ1374","FJ1375","FJ1379","FJ1382","FJ2554","FJ2555","FJ2556","FJ2557","FJ2558","FJ2741","FJ2748","FJ2753","FJ2760","FJ2761","FJ2762","FJ2766","FJ2768","FJ2772","FJ3199","FJ3201","FJ3263","FJ3265","FJ3269","FJ3272","FJ3273","FJ3281","FJ3287","FJ3289","FJ3369","FJ3371","FJ3375","FJ3378","FJ3379","FJ3386","FJ3392","FJ3394","FJ3395"]}
reason: None

Relation / operator chain

RL:S:ST2:location:00:surface-landmark
status: UNRESOLVED | operator: surface_membership
operand IDs: LM:S:ST2:location:20-40
direction: None | source statement: None
reason: argument landmark not uniquely executable

RL:S:ST2:location:01:surface-landmark
status: RESOLVED | operator: surface_membership
operand IDs: LM:S:ST2:location:7-11
direction: None | source statement: S:ST2:location
reason: None

Dependency edges (projection of existing explicit IDs)

LM:S:ST2:location:20-40 -> RL:S:ST2:location:00:surface-landmark [operand_to_relation]

LM:S:ST2:location:7-11 -> RL:S:ST2:location:01:surface-landmark [operand_to_relation]

RL:S:ST2:location:01:surface-landmark -> S:ST2:location [relation_to_statement]

S:ST2:location -> ST2 [statement_to_point]

Recorded data order

Stage 2 output does not record timestamped per-operation runtime order. The following preserves exact serialized Stage 2 data order.

LM:S:ST2:location:20-40 -> LM:S:ST2:location:7-11 -> RL:S:ST2:location:00:surface-landmark -> RL:S:ST2:location:01:surface-landmark -> S:ST2:location -> ST2

Final semantic execution result

{"location_statement_status":"UNRESOLVED","primary_location_status":"UNRESOLVED"}

Blocking dependency

{"blocking_items":[{"landmark_class":"bone.opening","node_id":"LM:S:ST2:location:20-40","node_type":"landmark","reason":"no unique atlas name match","source_raw":"infraorbital foramen","status":"UNRESOLVED"}],"downstream_stop_items":["RL:S:ST2:location:00:surface-landmark"],"first_unresolved_dependency":{"landmark_class":"bone.opening","node_id":"LM:S:ST2:location:20-40","node_type":"landmark","reason":"no unique atlas name match","source_raw":"infraorbital foramen","status":"UNRESOLVED"},"upstream_resolved_dependencies":["RL:S:ST2:location:01:surface-landmark","LM:S:ST2:location:7-11"]}

Coordinates generated: 0 | Legacy C coordinate referenced: 0

- [] semantic decomposition correct
- [] operator semantics correct
- [] operand identity correct
- [] dependency structure correct
- [] unresolved/conditional handling correct
- [] ACCEPT
- [] REJECT

TE20 - UNRESOLVED

WHO primary Location

S:TE20:location: On the head, just superior to the auricular apex.

Semantic decomposition / operands

LM:S:TE20:location:29-48
source: the auricular apex
class: soft_tissue_feature | status: UNRESOLVED | executor: atlas_name_resolution
identity: {"kind":null}
reason: no unique atlas name match

LM:S:TE20:location:7-11
source: head
class: region.body_region | status: RESOLVED | executor: fma_mesh
identity: {"fma_id":"FMA7154","kind":"fma_concept","part_ids":["FJ1252","FJ1253","FJ1282","FJ1285","FJ1286","FJ1289","FJ1294","FJ1295","FJ1297","FJ1298","FJ1299","FJ1301","FJ1302","FJ1304","FJ1305","FJ1306","FJ1308","FJ1309","FJ1317","FJ1320","FJ1322","FJ1323","FJ1324","FJ1328","FJ1331","FJ1336","FJ1337","FJ1340","FJ1345","FJ1346","FJ1348","FJ1349","FJ1350","FJ1352","FJ1353","FJ1355","FJ1356","FJ1357","FJ1359","FJ1360","FJ1368","FJ1371","FJ1373","FJ1374","FJ1375","FJ1379","FJ1382","FJ1730","FJ1731","FJ1732","FJ1733","FJ1738","FJ1739","FJ1740","FJ1743","FJ1744","FJ1745","FJ1746","FJ1747","FJ1748","FJ1749","FJ1750","FJ1751","FJ1758","FJ1759","FJ1760","FJ1762","FJ1767","FJ1769","FJ1770","FJ1775","FJ1779","FJ1780","FJ1781","FJ1783","FJ1784","FJ1785","FJ1786","FJ1787","FJ1788","FJ1789","FJ1790","FJ1791","FJ1792","FJ1795","FJ1797","FJ1798","FJ1800","FJ1801","FJ1806","FJ1807","FJ1808","FJ1810","FJ1814","FJ1817","FJ1822","FJ1826","FJ1828","FJ1830","FJ1831","FJ1833","FJ1834","FJ1835","FJ1836","FJ1841","FJ1842","FJ2554","FJ2555","FJ2556","FJ2557","FJ2558","FJ2741","FJ2748","FJ2753","FJ2760","FJ2761","FJ2762","FJ2766","FJ2768","FJ2772","FJ3199","FJ3200","FJ3201","FJ3263","FJ3265","FJ3269","FJ3272","FJ3273","FJ3274","FJ3281","FJ3287","FJ3289","FJ3309","FJ3369","FJ3371","FJ3375","FJ3378","FJ3379","FJ3380","FJ3386","FJ3392","FJ3394","FJ3395"]}
reason: None

Relation / operator chain

RL:S:TE20:location:00:relative-to
status: UNRESOLVED | operator: directional_relation
operand IDs: LM:S:TE20:location:29-48
direction: None | source statement: None
reason: argument landmark not uniquely executable

RL:S:TE20:location:01:surface-landmark
status: RESOLVED | operator: surface_membership
operand IDs: LM:S:TE20:location:7-11
direction: None | source statement: S:TE20:location
reason: None

Conditions

{"branch_id":"branch_1","condition_id":"CD:S:TE20:note:1:00","condition_type":"body_position","executor":"condition_contract","reason":null,"source_span":{"char_end":29,"char_start":6,"condition_type":"body_position","role":null,"source_raw":"With the auricle folded"},"source_statement_id":"S:TE20:note:1","status":"RESOLVED"}

Dependency edges (projection of existing explicit IDs)

CD:S:TE20:note:1:00 -> S:TE20:note:1 [condition_to_statement]
LM:S:TE20:location:29-48 -> RL:S:TE20:location:00:relative-to [operand_to_relation]
LM:S:TE20:location:7-11 -> RL:S:TE20:location:01:surface-landmark [operand_to_relation]
RL:S:TE20:location:01:surface-landmark -> S:TE20:location [relation_to_statement]
S:TE20:location -> TE20 [statement_to_point]
S:TE20:note:1 -> TE20 [statement_to_point]

Recorded data order

Stage 2 output does not record timestamped per-operation runtime order. The following preserves exact serialized Stage 2 data order.

LM:S:TE20:location:29-48 -> LM:S:TE20:location:7-11 -> CD:S:TE20:note:1:00 -> RL:S:TE20:location:00:relative-to -> RL:S:TE20:location:01:surface-landmark -> S:TE20:location -> S:TE20:note:1 -> TE20

Final semantic execution result

{"location_statement_status":"UNRESOLVED","primary_location_status":"UNRESOLVED"}

Blocking dependency

{"blocking_items":[{"landmark_class":"soft_tissue_feature","node_id":"LM:S:TE20:location:29-48","node_type":"landmark","reason":"no unique atlas name match","source_raw":" the auricular apex","status":"UNRESOLVED"}],"downstream_stop_items":["RL:S:TE20:location:00:relative-to"],"first_unresolved_dependency":{"landmark_class":"soft_tissue_feature","node_id":"LM:S:TE20:location:29-48","node_type":"landmark","reason":"no unique atlas name match","source_raw":" the auricular apex","status":"UNRESOLVED"},"upstream_resolved_dependencies":["RL:S:TE20:location:01:surface-landmark","LM:S:TE20:location:7-11"]}

Coordinates generated: 0 | Legacy C coordinate referenced: 0

- ☐ semantic decomposition correct
- ☐ operator semantics correct
- ☐ operand identity correct
- ☐ dependency structure correct
- ☐ unresolved/conditional handling correct
- ☐ ACCEPT
- ☐ REJECT